Bridging the Gap in Exam Handwritten Text Recognition: Dataset, Benchmark, and Modeling

Runrui Li, Lin Zhu, *Member, IEEE*, and Hua Huang, *Senior Member, IEEE*

***Abstract*—Handwritten text recognition (HTR) in examination scenarios has gained increasing attention for its role in intelligent grading systems. However, existing studies have not systematically modeled the complex handwriting phenomena inherent in exam settings, hindering a comprehensive understanding of the recognition challenges and limitations of current methods. Specifically, handwriting artifacts pose significant challenges to recognition models in two complementary aspects: sequentially, they disrupt the reading order and lead to non-monotonic sequences, while visually, they distort character structures and induce attention drift. To enable systematic benchmarking of exam handwriting, we first construct BNU-Exam-HTR, a large-scale dataset of handwritten exam text, and establish BNU-Exam-Benchmark, a fine-grained evaluation framework defining 12 representative challenges observed in real exam handwriting. To overcome these challenges, we further propose EduOCR, a recognition model with a collaborative dual-branch decoder. The Sequential Symbol Module (SSM) uses autoregressive decoding to handle non-monotonic sequences, while the Permutation-Aware Prediction Head (PPH) simulates artifact perturbations to guide the shared encoder in distinguishing characters from noise, thus stabilizing attention and mitigating alignment errors. Extensive experiments show that EduOCR consistently outperforms state-of-the-art HTR models, OCR tools, and multimodal large language models across all 12 challenges, demonstrating superior robustness and adaptability.**

***Index Terms*—Handwritten text recognition, exam handwritten text recognition, benchmark dataset, optical character recognition.**

I. Introduction

HANDWRITTEN Text Recognition (HTR) has long attracted attention because of its broad applications in historical document analysis [1], [2], signature verification [3], [4], and everyday text understanding. In educational settings, accurate HTR plays a crucial role in building effective Intelligent Grading Systems (IGS) [5], [6], [7]. Among these scenarios, handwriting on exam papers presents unique challenges. Beyond diverse writing styles, exam-scene text contains complex artifacts such as erasures, insertions, and rewrites that alter text layout and reading order. These artifacts introduce irregularities that are uncommon in ordinary handwritten or scene text and therefore make recognition particularly difficult [8].

Several studies have explored exam-related datasets, such as SCUT-EPT [9] and TAL-OCR-ENG/CHN [10] (see Table I). However, these works mainly treat exam handwriting as generic text and evaluate overall recognition accuracy on randomly divided test sets, without modeling or analyzing the diverse handwriting phenomena unique to exam scenarios. Moreover, these datasets are limited in both scale and subject diversity, hindering a comprehensive evaluation of model robustness across handwriting variations. Collectively, these limitations prevent a fine-grained understanding of recognition challenges, model weaknesses, and the intrinsic complexity of exam handwriting.

To address this limitation, we construct **BNU-Exam-HTR**, a large-scale, multi-subject dataset of handwritten text collected from the exam papers of 24,910 Chinese middle-school students. It contains 322.8 K text-line images and 5.8 M characters. Based on this dataset, we further establish the **BNU-Exam-Benchmark**, which categorizes representative handwriting phenomena into 12 challenge types covering text-length variations, handwriting styles, layout disruptions, and diverse writing artifacts. This design enables fine-grained evaluation of model performance under different challenges. In addition, we refine evaluation metrics to reflect both the averaged performance across challenges and the overall accuracy under real-world distributions.

We evaluate representative recognition methods on the benchmark, including CTC-based models [11], [12], [13] and attention-based encoder-decoder models (EDTRs) [14], [15], [16], [17]. The results show that existing approaches achieve high accuracy on common text but their performance degrades sharply on complex exam text, particularly in the presence of handwriting artifacts such as erasure, supplement, and switching. These results reveal a critical limitation of current architectures and motivate the development of a more artifact-resilient recognition model. A closer analysis shows that artifacts degrade recognition from two complementary perspectives (see Fig. 1). From the sequence perspective, artifacts disrupt the consistency between the logical reading order and the physical writing order. Models such as CTC, constrained by monotonic alignment [18], struggle to capture such non-linear sequences. From the visual perspective, artifacts overlap with character regions and distort structural boundaries, which often leads attention-based models to suffer from attention drift and feature-character misalignment.

Received 7 November 2025; revised 16 July 2026; accepted 3 August 2026. Date of publication 6 August 2026; date of current version 5 October 2026. This work was supported by the National Natural Science Foundation of China under Contract 62437001. Recommended for acceptance by Xiang Bai. *(Corresponding author: Hua Huang.)*

The authors are with the School of Artificial Intelligence, Beijing Normal University, Beijing 100875, China, also with the Beijing Key Laboratory of Artificial Intelligence for Education, Beijing 100875, China, and also with the Engineering Research Center of Intelligent Technology and Educational Application, Ministry of Education, Beijing 100875, China (e-mail: huahuang@bnu.edu.cn).

The dataset and code will be released at https://github.com/BNU-ERC-ITEA/BNU-Exam-HTR.

Digital Object Identifier 10.1109/TPAMI.2026.3721731

0162-8828 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPREHENSIVE COMPARISON OF EXISTING HTR DATASETS. THE PROPOSED BNU-EXAM-HTR EXPANDS THE SCALE AND DIVERSITY OF EXAM HANDWRITING, OFFERING BROAD COVERAGE IN SUBJECTS, WRITERS, AND ARTIFACTS, AND ESTABLISHING A NEW BENCHMARK FOR EDUCATIONAL HTR RESEARCH

Dataset	Language	Field	Type	Writers	Size	Year
IAM [19]	English	General HTR/Education	text	400	9.2K text lines, 448.9K char in text.	2000
NIST19 [20]	English	General HTR	character	3600	814K char.	2019
TAL_OCR_ENG [10]	English	Education	text	–	10.0K text lines, 429.2K char in text.	2020
IMGUR5K [21]	English	General HTR	word	5305	230K words, 892K char in word.	2021
GNHK [22]	English	General HTR/Education	text	687	9.3K text lines, 172K char in text.	2021
HCL 2000/2020 [23]	Chinese	General HTR	character	1000	3.75M char.	2000/2020
CASIA-HWDB 1.0-1.2 [24]	Chinese	General HTR	character	1020	3.7M char.	2011
CASIA-HWDB 2.0-2.2 [24]	Chinese	General HTR	text	1019	52K text lines, 1.3M char in text.	2011
SCUT-EPT [9]	Chinese	Education	text	2986	50K text lines, 1.27M char in text.	2018
TAL_OCR_CHN [10]	Chinese	Education	text	30	11.7K text lines, 194.8K char in text.	2020
SCUT-HCCDoc [25]	Chinese	General HTR/Education	text	–	116K text lines, 1.15M char in text.	2020
BNU-Exam-HTR	-	Education	text	24910	322.8K text lines, 5.8M char in text.	2025
BNU-Exam-HTR-EN	English	Education	text	2976	46.7K text lines, 1.8M char in text.	2025
BNU-Exam-HTR-ZH	Chinese	Education	text	21934	276.1K text lines, 4.0M char in text.	2025



Fig. 1. Representative failure cases of mainstream HTR models on exam handwriting, highlighting the impact of handwriting artifacts. CTC-based methods (left) exhibit sequence inconsistencies and misaligned character predictions, while attention-based methods (right) suffer from attention drift and text repetition under visually complex artifacts.

Building on these insights, we further propose **EduOCR**, a recognition model tailored for examination scenarios with a collaborative dual-branch decoding architecture. The primary branch, called the *Sequential Symbol Module* (SSM), replaces CTC with autoregressive decoding to flexibly model irregular and non-monotonic sequences. The auxiliary branch, called the *Permutation-Aware Prediction Head* (PPH), simulates the permutation disturbances introduced by handwriting artifacts and guides the shared encoder to disentangle character patterns from artifact features. This dual-branch collaboration alleviates alignment failures and enhances robustness to visual corruption, enabling reliable long-range sequence modeling even under severe artifacts. Extensive experiments demonstrate that EduOCR consistently outperforms state-of-the-art HTR models, OCR tools, and LLMs/VLMs in examination scenarios, while also exhibiting strong generalization to other tasks.

In summary, the contributions of this paper are:

1) *BNU-Exam-HTR Dataset and BNU-Exam-Benchmark*: To address the limitations of existing HTR datasets for examination scenarios, we introduce a large-scale BNU-Exam-HTR dataset. Building upon this foundation, we further establish the BNU-Exam-Benchmark, a fine-grained evaluation framework focused on examination scenarios and grounded in real writing phenomena.
2) *EduOCR Model*: We propose EduOCR, a novel recognition model for examination papers featuring a synergistic dual-branch decoder. Experimental results demonstrate that EduOCR consistently outperforms mainstream HTR models, OCR tools, and multimodal large models in exam scenarios, while also achieving competitive performance in general text recognition tasks.

The remainder of this paper is organized as follows: Section II reviews related works in examination scenarios. Section III introduces the construction of BNU-Exam-HTR, BNU-Exam-Benchmark and training set. Section IV details the EduOCR architecture and design principle. Section V presents comprehensive experimental results, analysis and discussion. Finally, Section VI concludes the paper.

II. RELATED WORKS

A. Exam HTR Datasets

The field of HTR is a long-standing area of research, historically supported by several seminal datasets such as IAM [19] for English and RIMES [26] for French. These foundational datasets are instrumental in advancing models for cursive and structured handwriting, though they primarily feature relatively clean papers from a limited number of writers. To better address real-world complexity, subsequent research has focused on more challenging and specialized domains. In the context of Chinese educational documents, the SCUT-EPT dataset [9] is a pioneering effort. Its main contribution is the introduction of realistic challenges, including character erasures, text line supplements, character/phrase switchings, and noisy backgrounds. Linguistically, TAL-OCR [10] expands coverage by constructing exam-scene datasets in English (TAL-OCR-ENG) and Chinese (TAL-OCR-CHN). SCUT-HCCDoc [25] and GNHK [22] further broaden the domain by incorporating substantial student-note collections in addition to exam papers, thereby enriching educational-document diversity. Recent work has also studied struck-out handwritten word detection and restoration for descriptive answer evaluation [27], showing the importance of correction artifacts.

However, exam-scene HTR datasets still lack scale and diversity. Despite its pioneering role in introducing handwriting artifacts, SCUT-EPT is constrained by a limited scale (50 K text lines from 2,986 writers) and incomplete coverage of

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

artifact diversity. Other datasets also face notable limitations: TAL-OCR is even smaller in scale, comprising 10.0 K English and 11.7 K Chinese text lines, totaling about 21.7 K samples. The SCUT-HCCDoc exam-related subsets and GNHK contain 30.3 K and 9.3 K text line images respectively. This shortage of suitable resources critically hampers the development and rigorous evaluation of next-generation HTR models for these mission-critical applications. Our work explicitly targets this gap.

B. Exam HTR Evaluation

Traditional HTR evaluation relies on standard metrics: English/French typically report Character Error Rate (CER) and Word Error Rate (WER) [28], whereas Chinese often use Character Correct Rate (CR) and Accuracy Rate (AR) [29]. The ICDAR 2013 Chinese handwriting recognition competition [30] is a representative benchmark using AR for offline Chinese handwritten text recognition. Results are usually reported as test-set averages, offering a concise yet coarse indication of overall performance. A single average score may obscure model weaknesses in exam settings. A model may appear strong in overall performance while failing on critical cases such as irregular handwriting styles, layouts, and artifacts. SCUT-EPT highlights several critical cases but lacks stratified annotation, preventing targeted assessment of robustness on a per-challenge basis. Without fine-grained evaluation, it is difficult to expose failure modes or measure true progress in model robustness. Recently, large-scale benchmarks such as Union14M-Benchmark [31] and OCRBench [32] have emerged to provide more standardized evaluation. These efforts are valuable for assessing overall robustness and cross-domain generalization. However, examination scenarios lack a dedicated fine-grained benchmark that systematically captures the diversity and difficulty of real handwriting phenomena.

To address these limitations, we introduce BNU-Exam-Benchmark, which organizes real exam text lines from BNU-Exam-HTR into 12 annotated challenge types in both English and Chinese, reporting both per-type and overall performance. This phenomenon-aware evaluation design enables more transparent and diagnostic assessment, revealing where models fail in exam settings rather than merely how well they perform on average.

C. Exam HTR Methodologies

The methodologies applied to HTR in educational contexts largely mirror the evolution of the broader Scene Text Recognition (STR) field. Initial and still prevalent approaches utilize architectures based on Connectionist Temporal Classification (CTC) [11]. Models such as CRNN [12] and SVTR (Scene Text Recognition with a Single Visual Model) [13] fall into this category, prized for their high inference efficiency. Their effectiveness, however, is largely confined to regular, near-linear text layouts, as their inherent monotonic alignment assumption fails to model the non-linear spatial arrangements prevalent in authentic examination papers. To address the need for greater flexibility, attention-based encoder-decoder models represent the dominant paradigm [16], [33], [34], [35], [36]. Document-level methods such as DAN [37] further explore segmentation-free attention modeling, while recent Chinese HTR methods study segmentation-free recognition or unified segmentation-recognition frameworks [38], [39]. Recent OCR-oriented VLMs and document parsing models, including MonkeyOCR [40], [41], dots.ocr [42], dots.mocr [43], DeepSeek-OCR [44], [45], GLM-OCR [46], and HunyuanOCR [47], also extend OCR toward layout-aware parsing and structured document understanding. By generating text auto-regressively, these methods can learn complex sequential dependencies and handle variations in layout more effectively. Nonetheless, it is a widely recognized limitation that this approach is susceptible to attention drift, particularly when processing long texts or noisy backgrounds, which can lead to a loss of synchronization between the visual cues and the generated text.

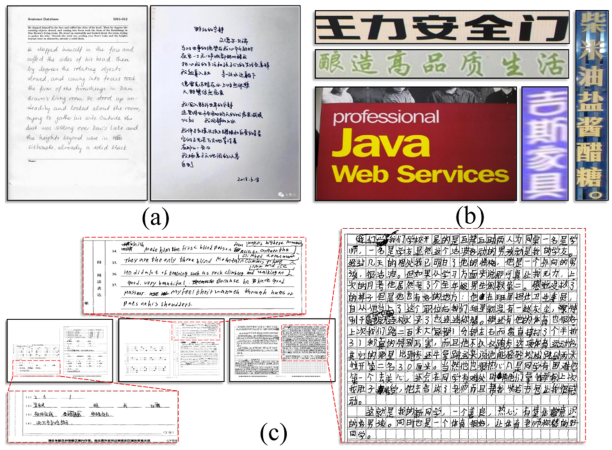


Fig. 2. Comparison of text recognition scenarios. (a) Traditional HTR focuses on well-formed handwriting with clean backgrounds and regular layouts. (b) STR handles in-the-wild text, challenged by variations in perspective, illumination, and background clutter. (c) Exam HTR presents unique challenges such as complex artifacts and irregular layouts.

Motivated by these limitations, we propose the EduOCR model tailored for examination scenarios. Our model features a collaborative dual-branch decoding structure that replaces the restrictive CTC assumption with an autoregressive mechanism and employs a permutation-aware auxiliary branch. This design is specifically engineered to ensure stable non-monotonic sequence modeling and robust feature alignment, offering a dedicated solution for exam HTR.

III. Dataset and Benchmark

A. Motivation and Overview

As shown in Fig. 2, unlike traditional HTR that targets clean scripts and STR that handles environmental distortions, exam handwriting poses its own distinct challenges: frequent corrections (erasure, supplement, replacement), severe layout irregularities (compressed text at line-end, unspaced text), and wide variation in text length. Examination scenarios also demand verbatim fidelity to the student handwriting, including spelling and grammatical errors, whereas other recognition tasks may

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

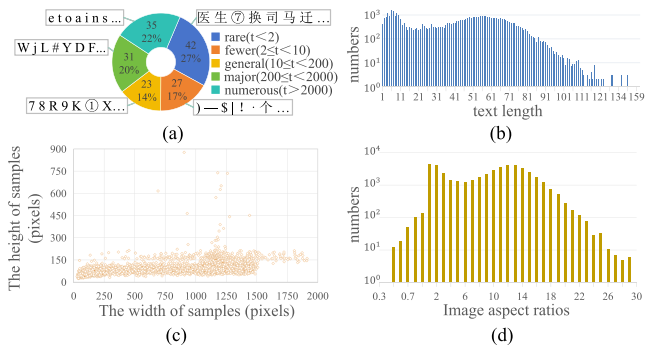


Fig. 3. Statistical analysis of BNU-Exam-HTR-EN. (a) Distribution of character frequencies, with representative sample characters. *t* denotes the number of occurrences for each character. (b) Distribution of text lengths. (c) Distribution of image dimensions. (d) Distribution of image aspect ratios. These statistics reveal the high diversity of handwriting styles and layout variations in exam scenarios.

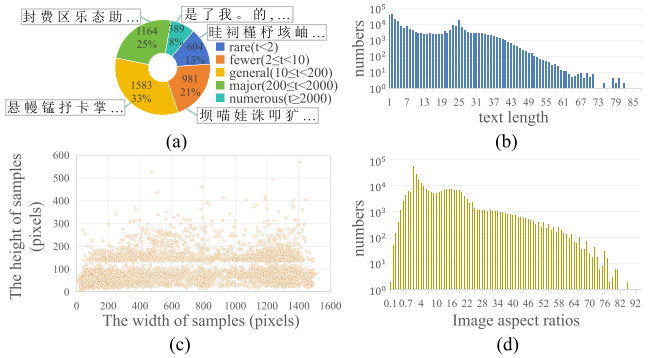


Fig. 4. Statistical analysis of BNU-Exam-HTR-ZH. (a) Distribution of character frequencies, with representative sample characters. *t* denotes the number of occurrences for each character. (b) Distribution of text lengths. (c) Distribution of image dimensions. (d) Distribution of image aspect ratios. The Chinese subset exhibits greater heterogeneity in character use and layout, highlighting language-specific complexity.

prioritize semantic accuracy. These characteristics are largely absent from existing datasets. To address this gap, we introduce BNU-Exam-HTR, a large-scale dataset capturing the unique complexities of student handwriting.

In the remainder of this section, we first describe the collection and annotation pipeline for BNU-Exam-HTR (Section III-B). Next, we construct the BNU-Exam-Benchmark for challenge-specific evaluation (Section III-C). We then present the composition of the training set (Section III-D) and design novel evaluation metrics tailored to the exam scenario (Section III-E).

B. Bnu-Exam-Htr

BNU-Exam-HTR comprises 24,910 answer-sheet images collected from Chinese middle and high school students across nine subjects (`Chinese`, `Mathematics`, `English`, `Politics`, `Geography`, `History`, `Physics`, `Chemistry`, and `Biology`). To protect privacy, all personal and exam-related metadata are removed (e.g., `student names`, `IDs`, `class`, and `school names`). The answer sheet images are cropped into text lines and finely annotated, excluding printed text and multiple-choice regions. Completely unreadable characters and entirely crossed-out lines are annotated with a special token "#". Finally, the dataset comprises 322.8 K text line images with 5.8 M characters. We further split the dataset by writing language into two subsets: **BNU-Exam-HTR-EN** containing 46.7 K English-subject images with 1.8 M characters, and **BNU-Exam-HTR-ZH** containing 276.1 K remaining-subject images with 4.0 M characters.

Fig. 3 presents a statistical analysis of the BNU-Exam-HTR-EN subset, highlighting its key data characteristics. **(a) Character Frequency Distribution:** The distribution is heavily long-tailed. We categorize characters into five tiers based on occurrence count (t): *rare* ($t < 2$), *fewer* ($2 \leq t < 10$), *general* ($10 \leq t < 200$), *major* ($200 \leq t < 2000$), and *numerous* ($t \geq 2000$). The two lowest-frequency tiers (*rare* and *fewer*) comprise 44% of the vocabulary, whereas the most frequent characters (e.g., 'e', 't', 'o') account for only 22% of the vocabulary. This severe skew is partly caused by students occasionally inserting Chinese symbols in English answers. **(b) Text Line Length Distribution:** The distribution of text line lengths (in characters) exhibits two distinct peaks, varying from single characters to over 100. The dual peaks correspond to short, phrase-level answers (peak at < 10 characters) and longer, sentence-level responses (peak around $50 - 60$ characters). **(c) Image Dimension Distribution:** The image heights are relatively consistent, mostly between 20 and 160pixels, forming a dense horizontal band. In contrast, widths vary widely, ranging from under 50 to nearly 1500pixels, reflecting diverse lengths and handwriting layouts. **(d) Aspect Ratio Distribution:** The distribution of aspect ratios (width/height) is also bimodal, with pronounced peaks near 1.0 (short, vertically compact answers) and 12.0 (long, horizontally extended sentences). The ratios span a wide range from 0.3 to over 25.

Fig. 4 presents the statistical analysis of the BNU-Exam-HTR-ZH subset, highlighting characteristics specific to Chinese handwriting. **(a) Character Frequency Distribution:** The distribution shows a broad but balanced character set. General-frequency characters account for the largest portion (33%), while only 13% are rare. Compared to English, the Chinese subset exhibits more consistent usage across mid- and high-frequency ranges. **(b) Text Line Length Distribution:** The distribution of text line lengths is distinctly bimodal. It features a sharp peak at fewer than 10 characters, corresponding to short, phrase-level answers, and a broader second peak centered around 25 characters, representing sentence-level responses. The overall distribution is concentrated below 60 characters, reflecting both the information density of Chinese script and the prevalence of structured answers in examinations. **(c) Image Dimension Distribution:** Text-line images exhibit relatively consistent heights, mostly between 20 and 200 pixels, while widths show moderate variation, with most samples distributed below 1500 pixels. This reflects stable vertical structure and varying horizontal lengths across responses. **(d) Aspect Ratio Distribution:** In contrast to the English data, the aspect ratio distribution for Chinese is more broadly dispersed, with a primary concentration of values between 0.2 and 60. This reflects the dual nature of

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

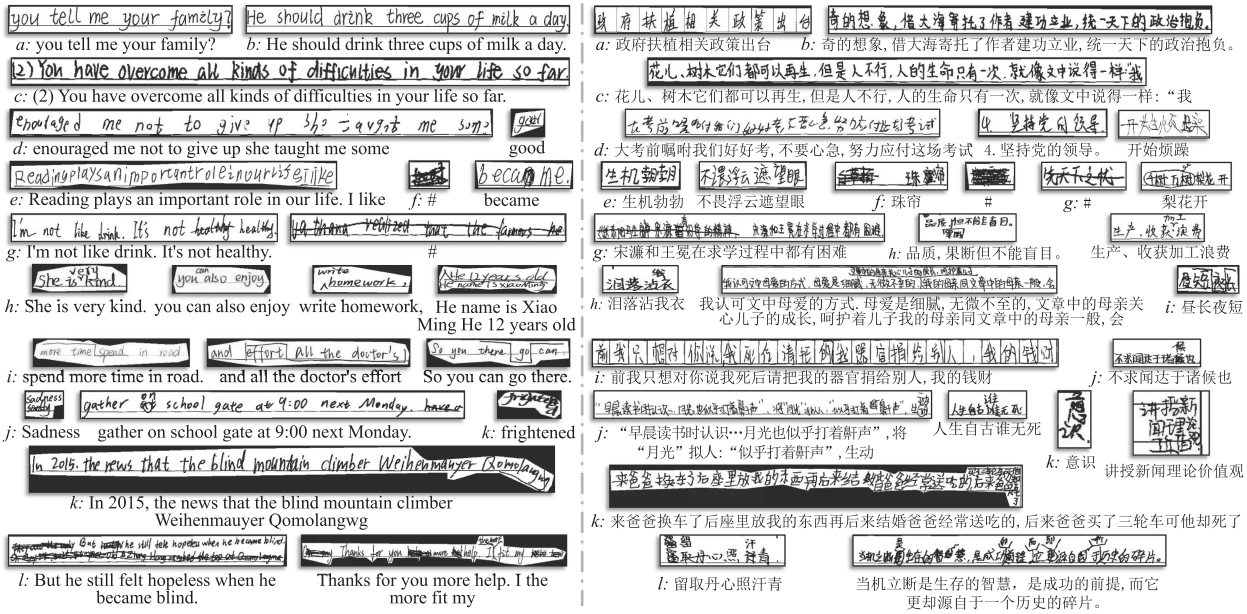


Fig. 5. Examples of the 12 challenge categories in the BNU-Exam-Benchmark. The figure displays BNU-Exam-Benchmark-EN (left) and BNU-Exam-Benchmark-ZH (right): (a-c) Challenges related to text length (`Common`, `Long`, `Ultra Long`); (d-e, k) Challenges related to handwriting quality and layout (`Low-quality Text`, `Compressed Text at Line End`, and the language-specific `Unspaced Text`/`Wrong Characters`); (f-j) Five typical correction-based challenges (`Hard/Soft Erasure`, `Supplement`, `Switching`, `Replacement`); and (l) `Multi-Type Challenges`, representing combinations of the above challenges.

TABLE II
STATISTICS OF THE BNU-EXAM-BENCHMARK SUBSETS. EACH SUBSET CONSISTS OF 12 CHALLENGE CATEGORIES: 11 ARE COMMON CROSS-LINGUAL CHALLENGES, AND ONE IS LANGUAGE-SPECIFIC. THE TABLE SHOWS THE SAMPLE NUMBERS FOR EACH CATEGORY

Challenge Type	English	Chinese
— 11 Common Challenges —		
common	6723	1403
long	1256	1121
ultra long	1414	1046
low-quality	1663	1098
hard erasure	1256	1289
soft erasure	1263	1196
supplement	1004	1084
switching	507	380
replacement	1073	1147
Compressed Text at Line End	1039	354
multi-type	1073	1057
— Language-Specific Challenge —		
unspaced (EN-specific)	887	–
wrong (ZH-specific)	–	1010
Total	**19173**	**12185**

Chinese handwriting in exam contexts: the frequent use of compact, single-character answers alongside longer, multi-character sentences.

The statistical analysis shows that BNU-Exam-HTR features a highly uneven character frequency distribution and extreme variability in text line lengths and aspect ratios. This comprehensive diversity makes it a highly challenging resource for handwritten text recognition research.

C. Bnu-Exam-Benchmark

To evaluate HTR models on realistic and challenging exam papers, we introduce **BNU-Exam-Benchmark**, a new benchmark derived from the BNU-Exam-HTR dataset. As shown in Table II, the benchmark is split into two language-specific subsets: English (BNU-Exam-Benchmark-EN) and Chinese (BNU-Exam-Benchmark-ZH). As illustrated in Fig. 5, each subset features 12 challenge categories. Eleven are cross-lingual, though their specific manifestations may vary, while the final category is language-specific and tailored to the unique features of each script. The detailed definitions of these challenges are described below:

Common, Long, and Ultra Long Text: The challenge posed by varying text lengths to recognition performance has recently become a research hotspot [48], [49]. Motivated by this, we define three challenge categories with text length (L) as the sole distinguishing variable: **common**, **long**, and **ultra long** (Fig. 5(a)–(c)). Samples differ only in length, with handwriting and backgrounds kept clean. Based on the real-world distributions observed in Figs. 3(b) and 4(b), we set the following thresholds for the English and Chinese scenarios: for English, the lengths are $L \leq 25$, $25 < L \leq 55$, and $L > 55$, respectively; for Chinese, they are $L \leq 15$, $15 < L \leq 35$, and $L > 35$.

Low-quality Text: As shown in Fig. 5(d), this category refers to text lines that remain annotatable but have low handwriting quality. It includes ambiguous character shapes, cursive or connected strokes, stroke omission caused by hurried writing, overlapping or overwritten characters, and broken strokes caused by writing or pen-related artifacts. Here, "low-quality" refers to degraded handwriting legibility rather than image acquisition quality. Completely unreadable or entirely crossed-out text is still annotated with the special token "#" when applicable.

Language-Specific Challenges: These challenges are specifically designed for English and Chinese, focusing on unique writing errors in each language. **Unspaced Text (English):** In exam scenarios, some students omit spaces, turning all words

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

into a single long string (as shown in Fig. 5(e, left)). This poses a significant challenge for word segmentation and recognition. **Wrong Characters (Chinese):** The intricate nature of Chinese characters leads students to write erroneous or non-existent forms (as shown in Fig. 5(e, right)). These errors result in out-of-vocabulary (OOV) characters, which strongly test model robustness.

Hard and Soft Erasures: Erasure is a common revision behavior in handwritten text. We classify erasure into two primary categories based on the legibility of the original script beneath the corrective marks. **Hard Erasure** refers to cases where the original text is obscured by heavy or repeated markings, rendering it mostly or completely illegible, as shown in Fig. 5(f). In contrast, **Soft Erasure** involves lighter or sparser markings, where the underlying original text remains largely legible, as shown in Fig. 5(g). As a special case, any line that is entirely crossed out is uniformly labeled with the special token "#".

Text Supplement: Another common revision behavior is inserting text between existing lines. The challenge lies in its diversity of forms, as shown in Fig. 5(h): Students may use various symbols, or no symbols at all, to indicate the insertion, and the added text can appear above or below the original line. This uncertainty in position and form poses a significant challenge for the correct segmentation and sequential reconstruction of text lines.

Text Switching: Text switching is another common correction method in papers, where symbols are used to transpose the order of characters or words within a line, as shown in Fig. 5(i). Two typical patterns are observed: swapping adjacent units ($AB \rightarrow BA$) and swapping units separated by an intervening element ($AOB \rightarrow BOA$). These markings pose significant challenges: models must not only recognize the different types of symbols but also correctly reorder the text sequence. This difficulty is exacerbated in Chinese, where symbols often overlap with background grid lines, making both detection and recognition more complex.

Text Replacement: Text replacement combines two behaviors: *Erasure* and a *Text Supplement:* The markings are diverse, ranging from simple combinations of erasure and supplement symbols to specialized replacement notations, as shown in Fig. 5(j). Thus, the model must not only detect erasure and supplement symbols but also accurately link the replacement text to its intended target.

Compressed Text at Line End: This challenge arises from "compressed" writing, which occurs at the end of a line when insufficient space is available. As shown in Fig. 5(k), the phenomenon takes several forms: (i) extremely cramped character spacing; (ii) text line drift upward or downward; and (iii) overflow writing, where characters that do not fit are placed above or below the line end. These irregular layouts and character deformations severely affect the accuracy and robustness of recognition models.

Multi-Type Challenges: As shown in Fig. 5(l), we define multi-type challenges as text lines containing multiple behaviors, which compound to pose the most severe difficulty for recognition performance.

TABLE III
COMPOSITION OF TRAINING SET, DETAILING THE REAL AND SYNTHETIC SOURCES FOR EACH LANGUAGE

Language	Dataset	Size
English	BNU-Exam-HTR-EN	27K
	TAL-OCR-ENG [10]	10K
	GNHK (line) [22]	4.6K
	GNHK (word) [22]	31.2K
	IAM [19]	6.5K
	IMGUR5K (word) [21]	157.9K
	Synthetic data	60K
	Total (EN)	**296K**
Chinese	BNU-Exam-HTR-ZH	264K
	CASIA-HWDB2.x [24]	52.2K
	TAL-OCR-CHN-train [10]	10.4K
	SCUT-EPT-train [9]	40K
	BCTR-hw (train) [50]	74.6K
	BCTR-hw (val) [50]	18.6K
	Synthetic data	60K
	Total (ZH)	**520K**

D. Training Set

Constructing a robust HTR model for exam scenarios requires extensive data diversity and scale. Following standard practice in the Scene Text Recognition (STR) community, where real-world models often rely on aggregating multiple large-scale datasets and employing synthetic data for robustness, we construct a comprehensive mixed training set (see Table III). This strategy is essential for three reasons: 1) It mitigates long-tail character distribution in our new dataset. 2) It provides diverse handwriting styles to enhance model generalization. 3) It allows for the targeted augmentation of specific, scarce writing artifacts via synthetic samples, which are critical for the BNU-Exam-Benchmark challenges. To ensure fair benchmarking, all public test sets are excluded. The training set is also split into English and Chinese subsets.

The English training set of 296 K samples is based on 27 K in-domain data from `BNU-Exam-HTR-EN`. To enhance stylistic diversity, we incorporate several public datasets, including both line-level (e.g., `TAL-OCR-ENG`, `IAM`) and word-level (e.g., `GNHK (word)`, `IMGUR5K`). We generate 60 K synthetic samples to mitigate the long-tail problem. These samples are evenly distributed (20 K each) across three categories: *Text Supplement*, generated from our 104 predefined insertion templates; *Text Switching*, created from our 126 predefined transposition templates; and *Line-End Compression*, which was synthesized by programmatically applying deformations to normal text lines. The template-based synthesis process for the former two categories is shown in Fig. 6. For *Text Supplement* and *Text Switching*, the templates are manually prepared from real exam samples. We first collect samples containing supplement or switching artifacts, erase the original text while preserving these artifacts, and annotate key regions. A supplement template contains the inserted-text region and its anchor position, while a switching template contains two or three candidate regions corresponding to $AB \rightarrow BA$ or $AOB \rightarrow BOA$. During synthesis, real text patches with similar region sizes are sampled from the training set and pasted into these regions to avoid large deformation; labels follow the corresponding supplement or switching rule.

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

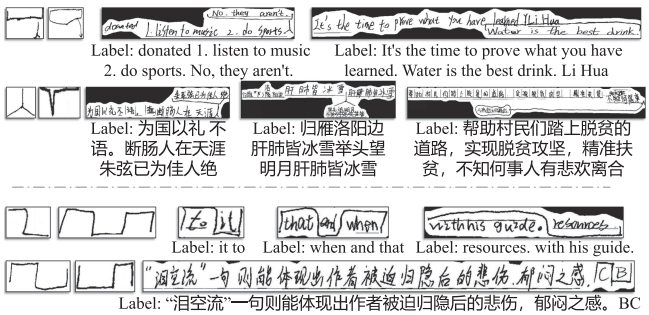


Fig. 6. Representative examples of template-based synthesis. (Top) Templates and samples for *Text Supplement*. (Bottom) Templates and samples for *Text Switching*.

Similarly, the Chinese training set consists of 520 K samples, primarily built upon 264 K in-domain samples from `BNU-Exam-HTR-ZH`. We enriched this set with data from public benchmarks, including `CASIA-HWDB2.x`, `TAL-OCR-CHN`, `SCUT-EPT`, and `BCTR-hw`. Complementing the real data, we include an additional 60 K synthetic samples with the same composition and rationale as the English set, targeting *Text Supplement*, *Text Switching*, and *Line-End Compression* to ensure the model is sufficiently exposed to these critical cases during training.

E. Evaluation Metrics

In this subsection, we refine the metrics for evaluating models in exam settings. We use PNAcc to measure sentence-level correctness after punctuation normalization, 1-NED to measure character-level similarity, and SMean/SWeight to summarize performance over the 12 challenge categories.

Punctuation-Normalized Accuracy (PNAcc): For exam HTR tasks, minor inconsistencies in punctuation usage (e.g., full-width vs. half-width symbols such as $'\circ'$ and $'.'$) are often semantically irrelevant and should not be penalized. Therefore, we introduce **Punctuation-Normalized Accuracy (PNAcc)** as a stricter yet reasonable sentence-level metric. A sample is correct only if the entire predicted text line perfectly matches the ground truth after both are normalized for such punctuation variants. It is computed as follows:

$$\text{PNAcc}_i = \begin{cases} 1, & \text{if Norm}(P_i) = \text{Norm}(G_i) \\ 0, & \text{otherwise} \end{cases}, \tag{1}$$

where P_i and G_i are the predicted and ground-truth sequences for a given sample i. $\text{Norm}(\cdot)$ is the function that standardizes equivalent punctuation marks.

One Minus Normalized Edit Distance (1-NED) [51]: This complementary metric provides a more granular, character-level analysis of model performance:

$$\text{1-NED}(P_i, G_i) = 1 - \frac{\text{EditDist}(P_i, G_i)}{\max(|G_i|, 1)}, \tag{2}$$

where $\text{EditDist}(P_i, G_i)$ is the Levenshtein distance between the predicted sequence P_i and the ground-truth G_i. A higher score indicates a better character-level match.

TABLE IV
EMPIRICAL FREQUENCY WEIGHTS (w_k) FOR 12 CHALLENGE CATEGORIES, ESTIMATED FROM FREQUENCY STATISTICS FOR EACH LANGUAGE. THESE WEIGHTS ARE USED TO COMPUTE THE S_{WEIGHT} SCORE FOR EVALUATION

Challenge Type	English	Chinese
common	0.252	0.496
long	0.159	0.176
ultra long	0.227	0.021
low-quality	0.029	0.068
unspaced/wrong	0.021	0.008
hard erasure	0.133	0.133
soft erasure	0.085	0.042
supplement	0.017	0.011
switching	0.007	0.002
replacement	0.030	0.026
line-end compressed	0.007	0.004
multi-type	0.033	0.013

We aggregate the per-challenge average scores ($\bar{S}_k$) into two final metrics. To measure general robustness across all K challenges, we compute the **Mean Average** (S_{Mean}). To provide a more realistic assessment, we compute the **Weight Average** (S_{Weight}). They are defined as:

$$S_{\text{Mean}} = \frac{1}{K}\sum_{k=1}^{K} \bar{S}_k, S_{\text{Weight}} = \sum_{k=1}^{K} w_k \cdot \bar{S}_k, \text{s.t.} \sum_{k=1}^{K} w_k = 1, \tag{3}$$

where weights w_k are empirical frequencies of challenges, estimated from 1,000 random samples (see Table IV).

IV. Proposed Model

A. Architecture Motivation and Overview

Existing HTR methods exhibit marked performance degradation in examination scenarios. We attribute this to two inherently conflicting challenges stemming from the domain's complexity: (i) **Visual dimension.** The model must extract robust features from severely degraded visual inputs, including distorted character shapes caused by hurried handwriting, cursive ligatures, and overlaps, while disentangling structural interference from artifact symbols. (ii) **Sequential dimension.** The model must faithfully reconstruct the writer's intended reading order while calibrating overly strong language priors to avoid "semantic-correction" hallucinations. Conventional decoding paradigms struggle to address both challenges simultaneously. CTC-based models are fundamentally constrained by their monotonic-alignment and conditional-independence assumptions, making them ill-suited to model nonlinear spatial layouts. Attention-based encoder-decoder text recognizers (EDTRs) are prone to attention drift during spatial alignment.

Based on the analysis, we propose **EduOCR**, a novel recognition model tailored to examination scenarios. As shown in Fig. 7, EduOCR first processes an input image $X \in \mathbb{R}^{H\times W\times 3}$ through a shared visual encoder (Section IV-B) to extract a feature map $F \in \mathbb{R}^{\frac{H}{16}\times\frac{W}{4}\times C_2}$. Next, F is fed to a dual-branch decoder comprising the Permutation-Aware Prediction Head (PPH) and the Sequential Symbol Module (SSM) (Section IV-C). By supervising the encoder through multiple noise-perturbed queries, PPH encourages the model to extract robust and discriminative

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

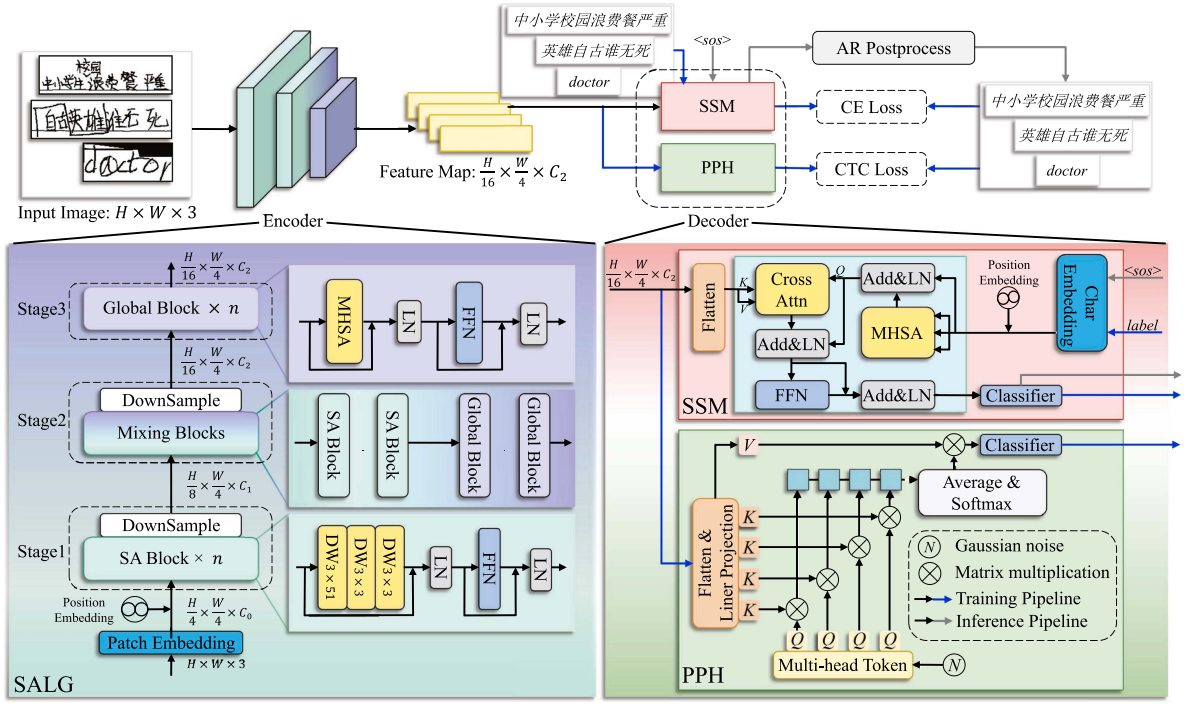


Fig. 7. Overall architecture of EduOCR. It comprises a shared visual encoder, a Symbol-Aware Local and Global encoder (SALG), and a dual-branch decoder: a primary Sequential Symbol Module (SSM) for autoregressive sequence generation, and an auxiliary Permutation-Aware Prediction Head (PPH) that provides CTC-supervised regularization for the shared encoder. The model is jointly optimized using CTC and CE losses during training. Only SSM is retained in inference.

features from visually degraded and structurally disordered input images. This mechanism acts as an implicit regularization, enforcing the encoder to focus on stable visual cues while ignoring spurious variations. SSM addresses the sequential dimension: leveraging PPH-regularized features, it explicitly models long-range inter-character dependencies via autoregressive decoding, thereby reconstructing the intended reading order and resolving ambiguities while accommodating irregular layouts. This architecture decouples robust feature learning (via PPH) from precise sequence generation (via SSM), enabling the model to address spatial and sequential challenges concurrently. The entire model is trained end-to-end with a joint objective function combining CTC and CE losses (Section IV-D). During inference, we retain only SSM.

B. Encoder Architecture

The encoder's primary objective is to extract robust visual features that are invariant to handwriting degradation while being sensitive to artifact symbols and varying aspect ratios, which are prevalent in exam papers. Building upon the SVTR (Scene Text Recognition with a Single Visual Model) architecture [13], the EduOCR encoder, termed Symbol-Aware Local and Global (SALG), introduces a novel **Symbol Awareness Block (SAB)** to handle some artifact symbols with extreme aspect ratios. This module augments each local block with a depthwise-separable convolution that expands the receptive field exclusively along the horizontal dimension, as shown in the bottom-left panel of Fig. 7. We first apply an overlapping patch embedding layer to convert the input image into a $\frac{H}{4} \times \frac{W}{4}$ sequence of C_0-dimensional tokens and add position embeddings. Similar to SVTR, our backbone consists of three stages for feature extraction, with each stage containing N_i blocks. We employ SAB to replace local blocks in Stage 1 and the beginning of Stage 2. For the remainder of Stage 2 and Stage 3, we use Global Blocks to capture global information. Downsample operations are performed after both Stage 1 and Stage 2, which halves feature map height and increases the number of channels while maintaining a constant width. Finally, we obtain the feature map $F \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{4} \times C_2}$.

C. Decoder Architecture

The key to handling the exam HTR challenges lies in decoupling feature learning from sequence modeling. The proposed collaborative dual-branch decoder is engineered to simultaneously achieve two conflicting goals: acquiring permutation-robust visual features and generating non-monotonic text sequences. As shown in the bottom-right panel of Fig. 7, the architecture comprises two synergistic components: the Permutation-Aware Prediction Head (PPH) and the Sequential Symbol Module (SSM).

Permutation-Aware Prediction Head (PPH):

We introduce a novel PPH, which drives the model to learn robust features from visually degraded and irregularly laid-out inputs by explicitly modeling two perturbation types. First, we introduce learnable base tokens $Q_b \in \mathbb{R}^{N \times D}$, which serve as a prototype query for character identities. N denotes the maximum sequence length, and D denotes the query and key feature dimension. We set $D = C_2$ to match the channel dimension of the final encoder feature. To handle non-linear spatial layouts,

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

we add learnable permutation bias $Q_p \in \mathbb{R}^{N\times D}$ to Q_b. This bias is designed to model and compensate for feature disturbances caused by spatial permutations. During training, Q_p is optimized to shift the queries, enabling it to flexibly locate targets at different positions on the feature map and thereby endowing the model with the ability to perceive and adapt to non-linear character layouts. Concurrently, to simulate morphological degradations and deformations, we inject random gaussian noise $\epsilon_k \sim \mathcal{N}(0, \sigma^2\mathbf{I})$ into Q_b. The process of averaging the results from multiple noisy queries effectively acts as a denoising step. It filters out interference caused by degradation while reinforcing essential features that remain stable under perturbation. This compels the model to learn feature representations that are less sensitive to morphological variations. The final composite query vector is formulated as:

$$Q = Q_b + Q_p + \epsilon_k. \tag{4}$$

We extract key (K) for matching and value (V) for information aggregation from 2-D feature map $F \in \mathbb{R}^{H'\times W'\times C_2}$. First, we flatten F into 1-D sequence $F_{seq} \in \mathbb{R}^{(H'W')\times C_2}$. Subsequently, we use a shared linear projection layer W_{kv} to simultaneously map each feature vector in the sequence into two parts, the key and the value:

$$[K, V] = Linear(F_{seq}) = F_{seq}W_{kv}. \tag{5}$$

We compute raw attention scores for Q and K and average them. We then apply the Softmax function to convert the result into a normalized attention weight matrix α:

$$\alpha = \text{Softmax}\left(\frac{1}{n}\sum_{i=1}^{n}\frac{Q_i K^T}{\sqrt{d}}\right), \tag{6}$$

where n is the number of noisy perturbations, Q_i is the i-th noisy query, and d is the dimension used as a scaling factor. Finally, we use α to perform a weighted aggregation of V:

$$F_{\text{PPH}} = \alpha V. \tag{7}$$

Sequential Symbol Module (SSM): The architecture of the SSM is, by design, conventional. Its strong performance is not inherent to its structure but is a direct consequence of the features it receives from the shared encoder. The primary component consists of n standard Transformer decoder layers [52]. At step t, the SSM predicts current character y_t conditioned on both F and the previously generated tokens $y_{<t}$. The process begins by forming the input T_{emb} from char embedding and position embedding (PE):

$$T_{emb} = \text{Char Embedding}(y_{<t}) + \text{PE}(t). \tag{8}$$

T_{emb} is processed by a masked multi-head self-attention (MSA) layer, followed by a cross-attention (CA) that aligns the linguistic context with the visual evidence:

$$H'_{self} = \text{LN}\left(T_{emb} + \text{MSA}(T_{emb})\right), \tag{9}$$

$$H_{\text{cross}} = \text{LN}\left(H'_{self} + \text{CA}\left(\text{q} = H'_{self}, \text{k} = K_v, \text{v} = V_v\right)\right), \tag{10}$$

where LN denotes Layer Normalization [53]. The resulting features H_{cross} are then passed through a Feed-Forward Network (FFN), the final component of the decoder layer:

$$F_{SSM} = \text{LN}\left(H_{\text{cross}} + \text{FFN}(H_{\text{cross}})\right), \tag{11}$$

where the FFN consists of two linear layers with a ReLU activation [54]. The final feature F_{SSM} is then passed through a linear classifier to produce the output logits L_{SSM_t} for the current timestep. The demanding, non-autoregressive objective of PPH provides a powerful regularization signal, forcing the encoder to learn more robust and discriminative visual representations. SSM, in turn, leverages these high-quality features to achieve a significant improvement in recognition accuracy and fidelity, demonstrating the effectiveness of our synergistic training approach.

D. Loss Functions

The entire model is trained end-to-end via a joint training objective, $\mathcal{L}_{\text{total}}$, which is a weighted combination of the Connectionist Temporal Classification (CTC) loss from the PPH and the Cross-Entropy (CE) loss from the SSM:

$$\mathcal{L}_{\text{total}} = \lambda_1\mathcal{L}_{\text{CTC}}\left(L_{\text{PPH}}, y_{\text{true}}\right) + \lambda_2\mathcal{L}_{\text{CE}}\left(L_{SSM}, y_{\text{true}}\right), \tag{12}$$

where L_{PPH} and L_{SSM} are the output logits from the PPH and SSM branches, respectively. y_{true} is the ground-truth label corresponding to the input sample. Following SVTRv2 and considering that PPH serves as an auxiliary optimization branch, we set $\lambda_1 = 0.05$ and $\lambda_2 = 1$. During inference, only the SSM is utilized.

V. Experiments

A. Experiment Setting

We conduct experiments for both the English and Chinese HTR tasks. All models are trained on the respective composite training sets detailed in Section III-D. The models are then evaluated on our BNU-Exam-Benchmark, for which the English and Chinese subsets are described in Section III-C.

Training is performed using the AdamW optimizer [55] with a weight decay of 0.05. The learning rate (LR) is set to 6.5×10^{-4} and the batch size is set to 256. All images are resized to 32×512. We do not use SVTRv2's Multi-Size Resizing (MSR) in experiments; all methods follow the same fixed-size preprocessing for a fair comparison. A one-cycle learning rate scheduler [56] with 2- and 3-epoch linear warm-ups for English and Chinese, respectively, is adopted, with maximum text lengths of 100 and 60. Data augmentations, including perspective distortion, rotation, and motion blur, are randomly applied during training [57]. The evaluation metrics are PNAcc and 1-NED, with final performance reported as Mean Average (S_{Mean}) and Weight Average (S_{Weight}). The size of the character set (N_c) is set to 95 for English and 6996 for Chinese, with both including the space character. All models are trained with mixed-precision on 2 NVIDIA A800 GPUs.

B. Comparison With State-of-The-Art Models

To demonstrate the effectiveness of our proposed method, we conduct a comprehensive comparison between EduOCR and

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

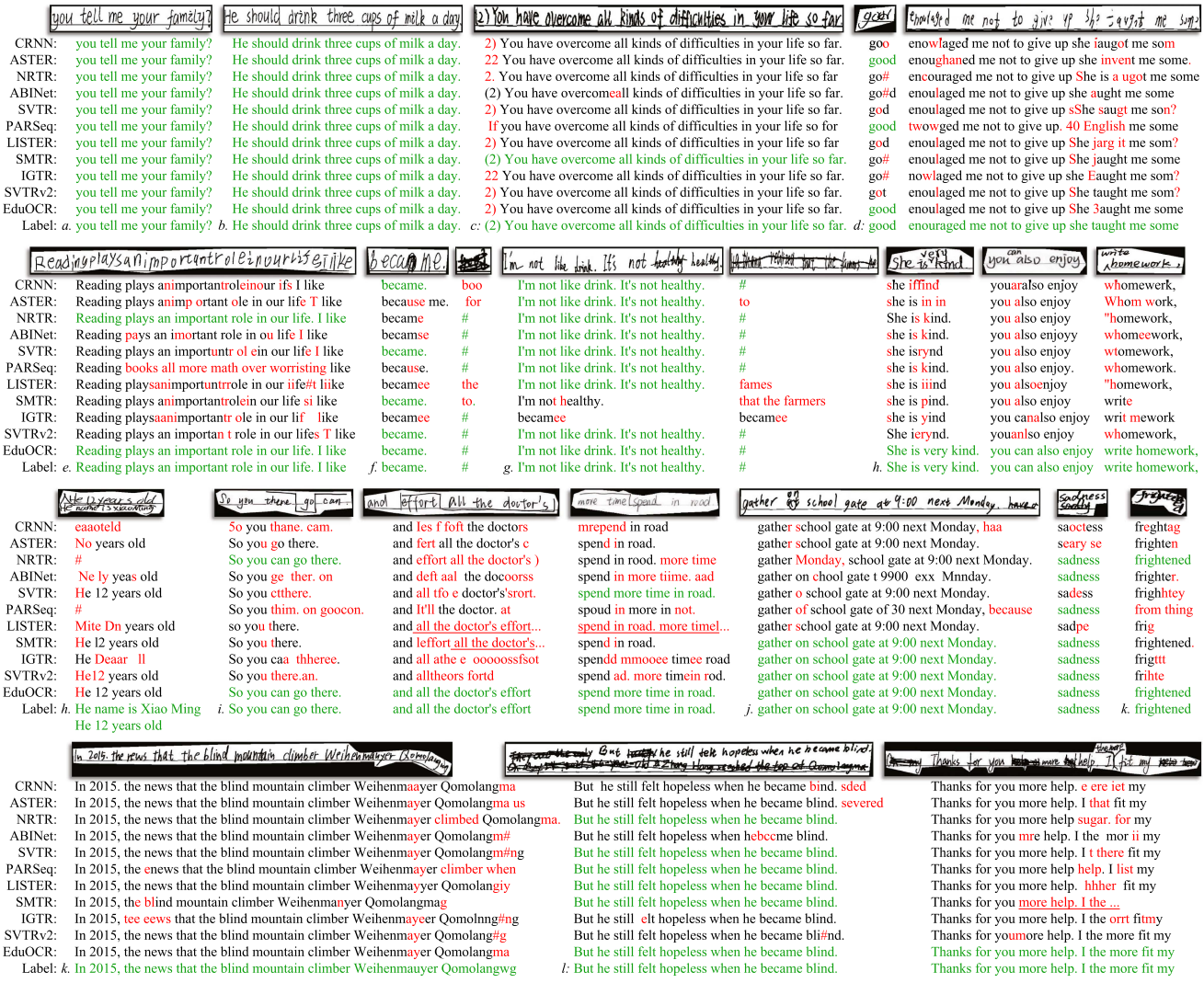


Fig. 8. Visual comparison of model performance on the BNU-Exam-Benchmark-EN dataset, using the examples from Fig. 5. Total line correct predictions are in green; errors are in red.

state-of-the-art (SOTA) HTR models: CRNN [12], ASTER [14], NRTR [15], ABINet [58], SVTR (Scene Text Recognition with a Single Visual Model) [13], PARSeq [17], LISTER [48], SMTR [49], IGTR [59], SVTRv2 [57], and SVTRv2-AR. SVTRv2-AR denotes an autoregressive variant of SVTRv2 that uses the Semantic Guidance Module (SGM) branch for decoding during inference instead of the default CTC branch. We evaluate on both the English and Chinese subsets of our BNU-Exam-Benchmark. For a fair comparison, all models are trained under the unified setting described in Section V-A.

1) Test on BNU-Exam-Benchmark-EN: Qualitative Experiments: As shown in Fig. 8, we visualize recognition results for all English examples from Fig. 5. Total line correct predictions are in green; errors are in red. Cases (a)–(c) test length-related challenges. Most models correctly recognize (a) `Common` and (b) `Long` text, but several fail on the (c) `Ultra Long`. This indicates that text length remains a significant challenge. In cases (d), (e), and (k), the irregular handwriting and layout cause more frequent recognition errors. These failures are mostly local: models misread ambiguous characters in (d) `Low-quality` and fail on word segmentation in (e) `Unspaced`, while the overall sentence structure is largely preserved and does not collapse. Cases (f)–(j) focus on exam artifacts. In these examples, CTC-based models often skip characters around the artifact positions, whereas EDTR models frequently repeat the same text segment. Particularly in (h) `Supplement` and (i) `Switching`, most existing models fail recognition, indicating that sequence-disrupting artifacts are especially challenging and the errors are no longer confined to a few characters but can affect the readability of the whole text. In case (l), `Multi-Type` challenges co-occur within a single sentence, which further increases the recognition difficulty for existing models. In contrast, EduOCR usually preserves the sentence structure and confines errors to a few local character substitutions, avoiding both truncation and repeated results. Overall, among the 23 examples, EduOCR correctly recognizes 19, whereas NRTR recognizes 10 and SVTRv2 and PARSeq each recognize 8, indicating that most

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

TABLE V
PERFORMANCE COMPARISON ON THE BNU-EXAM-BENCHMARK-EN. FOR EACH METHOD, THE TOP AND BOTTOM ROWS SHOW PNACC (%) AND 1-NED (%) SCORES, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE. LAST COLUMN SHOWS MODEL PARAMETERS (PARAMS, IN MILLIONS). ASTER∗ MEANS ASTER WITHOUT TPS

Method	comm.	long	ultra	low-qual.	unsp.	hard era.	soft era.	suppl.	switch.	repl.	comp.	multi	S_{Mean}	S_{Weight}	Params
CRNN	81.05	79.37	47.31	49.66	17.58	27.82	30.40	0.89	0.00	9.78	0.67	3.42	29.00	52.30	16.19
(TPAMI 2016)	97.81	99.53	98.71	96.18	95.48	88.86	84.47	81.28	62.46	88.45	78.18	83.82	82.82	94.45	
ASTER*	93.42	84.79	58.48	63.68	37.31	40.86	42.91	3.38	4.14	23.48	4.33	8.06	38.74	63.09	17.18
(TPAMI 2019)	99.70	99.68	98.86	96.79	94.19	86.05	80.65	80.12	67.25	89.51	79.84	83.97	88.00	94.34	
NRTR	95.74	82.48	59.12	72.88	45.99	54.30	51.14	6.87	2.76	23.57	8.27	11.21	42.86	66.58	44.25
(ICDAR 2019)	99.68	99.60	99.61	98.07	96.84	92.52	87.44	78.42	63.49	88.40	78.00	82.79	88.66	95.89	
ABINet	92.77	48.72	47.10	48.82	17.36	38.02	28.50	3.78	2.95	20.41	3.94	4.07	29.70	51.93	36.85
(CVPR 2021)	99.44	98.56	98.63	96.76	95.00	91.78	87.22	85.15	68.79	93.68	85.46	87.28	90.64	95.75	
SVTR	94.05	87.73	61.03	65.42	31.34	44.34	48.77	11.65	11.43	33.08	3.84	11.30	42.00	65.77	25.40
(IJCAI 2022)	99.53	99.83	99.29	98.25	97.60	94.37	92.13	89.11	80.90	95.28	86.48	89.66	93.54	97.27	
PARSeq	93.27	54.69	19.23	42.99	15.21	38.41	32.62	3.28	0.59	13.04	4.04	1.66	26.59	46.54	23.97
(ECCV 2022)	99.49	96.96	91.50	93.03	81.10	86.66	83.24	72.80	53.71	82.08	68.91	68.78	81.52	91.08	
LISTER	92.25	84.87	51.41	54.53	27.05	39.52	42.75	1.09	4.33	13.32	0.96	5.93	34.83	60.10	48.61
(ICCV 2023)	99.27	99.75	98.94	96.93	96.11	88.40	81.28	81.22	67.56	88.75	80.61	83.80	88.55	94.67	
SMTR	94.71	73.24	41.30	58.26	29.42	47.35	41.33	10.85	2.56	32.05	6.54	9.91	37.29	58.54	15.81
(AAAI 2025)	99.57	90.73	82.46	90.36	84.29	83.43	74.81	74.01	58.76	81.19	75.43	70.35	80.45	87.03	
IGTR	92.39	88.37	63.57	64.22	23.56	44.03	46.47	4.98	1.97	30.09	3.36	8.15	39.26	65.22	24.65
(TPAMI 2025)	99.25	99.70	99.00	97.27	93.62	91.19	85.19	84.70	69.74	93.80	87.22	85.54	90.52	95.66	
SVTRv2	96.38	90.84	70.57	70.83	49.26	62.21	65.24	19.52	15.38	47.25	10.29	17.23	51.25	74.72	21.02
(ICCV 2025)	99.41	99.74	99.35	98.39	97.72	95.50	92.29	89.76	82.63	96.99	88.24	90.03	94.17	97.63	
SVTRv2-AR	96.23	74.28	41.51	61.51	36.07	58.18	58.98	27.19	18.34	41.75	17.42	14.92	45.53	63.78	21.02
(ICCV 2025)	99.34	90.63	82.16	90.68	84.75	89.38	85.75	79.98	75.32	84.23	79.97	74.25	84.70	89.27	
EduOCR-CTC	96.10	90.52	69.51	70.41	44.98	59.60	62.23	20.61	22.09	44.45	6.73	16.12	50.28	73.53	25.58
	99.39	99.75	99.33	98.31	97.10	94.34	92.23	89.94	86.94	96.37	86.45	89.63	94.15	97.41	
EduOCR	96.86	90.60	74.25	74.74	56.48	64.42	70.38	39.14	32.34	56.75	18.57	27.06	58.47	77.64	25.58
	99.50	99.73	99.46	98.53	97.90	95.58	94.92	92.58	87.01	97.14	88.55	91.76	95.22	98.03	

difficult cases that defeat existing CTC and EDTR architectures are successfully handled by EduOCR.

Quantitative Experiments: As shown in Table V, we evaluate all methods on BNU-Exam-Benchmark-EN, where the top and bottom rows report PNAcc and 1-NED, respectively. ASTER∗ denotes ASTER without the Thin Plate Spline (TPS) [60] rectification module, since exam text lines are roughly horizontal. The results show a clear mismatch between character-level similarity and line-level correctness: several baselines maintain high 1-NED but low PNAcc, such as CRNN on `Supplement` (0.89% PNAcc vs. 81.28% 1-NED), indicating that minor local errors can still break exact exam transcription. SVTRv2-AR improves several artifact categories over SVTRv2, but its overall score remains lower. EduOCR-CTC obtains 50.28% S_{Mean} and 73.53% S_{Weight}, showing the limitation of the CTC inference branch on artifact-heavy cases. In contrast, EduOCR achieves the best overall performance, with 58.47% S_{Mean} and 77.64% S_{Weight}, outperforming SVTRv2 by 7.22% and 2.92%, respectively. It ranks first in PNAcc for 11 of 12 categories and first in 1-NED for 9 categories. The gain is especially clear on order-disrupting artifacts, such as `Supplement` (39.14% vs. 19.52% of SVTRv2) and `Switching` (32.34% vs. 15.38%), indicating stronger robustness to structural disorders and non-monotonic layouts.

2) Test on BNU-Exam-Benchmark-ZH: Qualitative Experiments: As shown in Fig. 9, we also visualize recognition results for all Chinese samples from Fig. 5. Compared with the English cases, recognition errors are more frequent in the Chinese examples, reflecting the extra difficulty caused by complex strokes and numerous characters in Chinese. Several models produce recognition errors even on the relatively easy (a) `Common` and (b) `Long` challenges. Irregular handwriting and layout challenges have a stronger impact in Chinese. The complexity of Chinese characters, combined with blurred or non-standard writing, makes character recognition errors more frequent. In addition, Chinese text lengths are on average shorter than English ones (see Figs. 3(b) and 4(b)), so these local errors, which might remain isolated in English, are much more likely to destroy the overall sentence structure and corrupt the intended meaning in Chinese. Cases (f)–(j) focus on exam artifacts. In these examples, CTC-based models often skip characters around the artifact positions, whereas EDTR models frequently repeat the same text segment. Particularly in (h) `Supplement` and (i) `Switching`, most existing models fail recognition, indicating that sequence-disrupting artifacts are especially challenging and the errors are no longer confined to a few characters but can affect the readability of the whole text. Overall, EduOCR is the most robust model on these Chinese examples. Among the 27 samples, it correctly recognizes 18, outperforming the next-best model, SMTR (15 correct), as well as SVTRv2 and IGTR (12 correct each), which confirms its advantage on Chinese exam handwriting.

Quantitative Experiments: As shown in Table VI, the quantitative results confirm the heightened difficulty of the Chinese benchmark. Compared with the English benchmark, the Chinese benchmark yields consistently lower PNAcc S_{Mean} scores across all methods. In addition, the mismatch between high 1-NED and near-zero PNAcc is even more pronounced on Chinese text lines. SVTRv2-AR slightly improves over SVTRv2 in the overall scores (38.11%/70.62% vs. 37.31%/69.40%), and also improves artifact categories such as `Supplement` (20.97% vs. 17.56%) and `Compressed at line end` (11.01% vs. 7.34%). EduOCR-CTC obtains 36.69% S_{Mean} and 69.23% S_{Weight}, which is close to the CTC-based SVTRv2 but remains

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

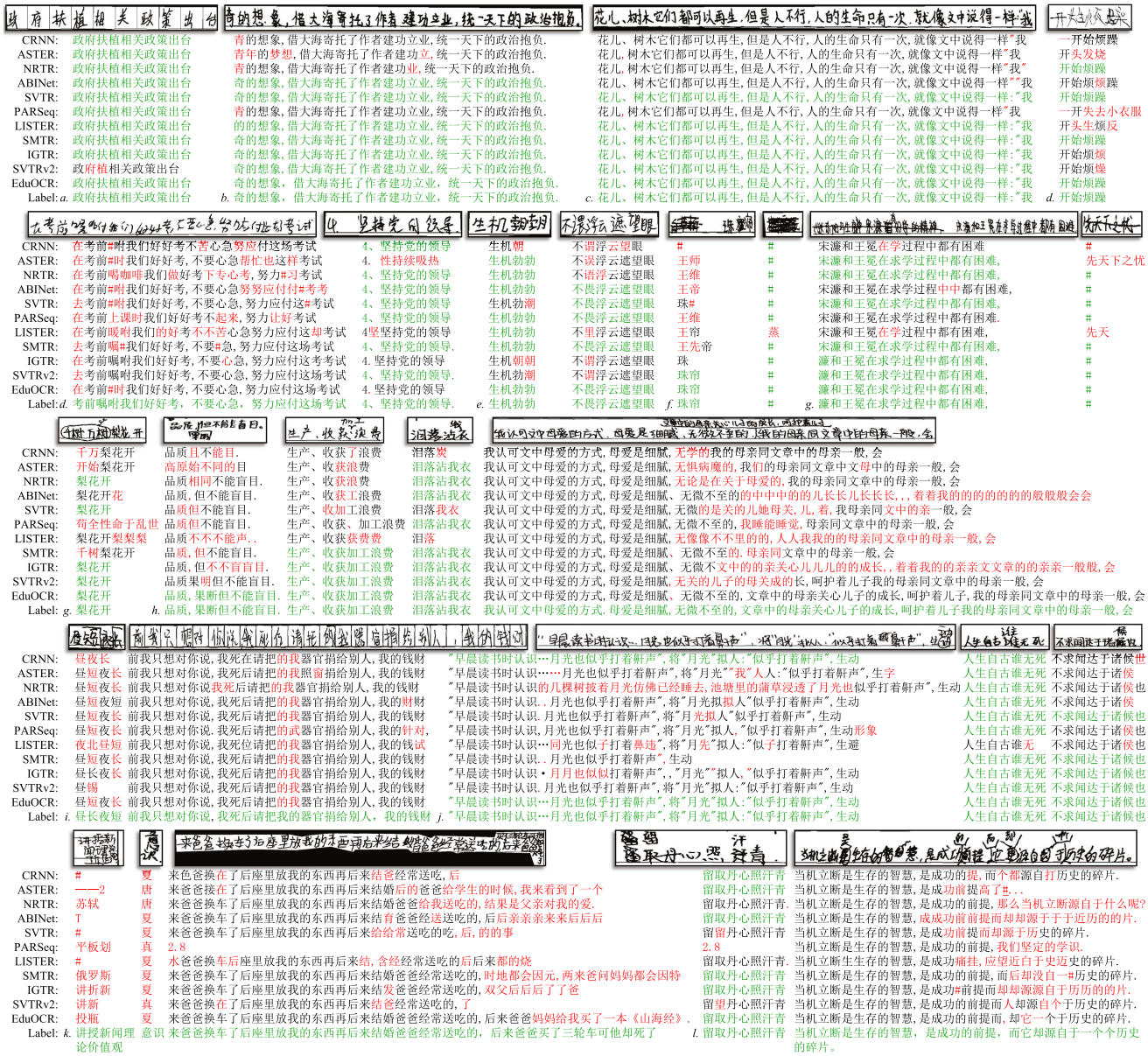


Fig. 9. Visual comparison of model performance on the BNU-Exam-Benchmark-ZH dataset, using the examples from Fig. 5. Total line correct predictions are in green; errors are in red.

below the autoregressive EduOCR. EduOCR similarly achieves the best performance on the Chinese benchmark with S_{Mean} of **43.18%** and S_{Weight} of **73.99%**, outperforming SVTRv2-AR (38.11% of S_{Mean}, 70.62% of S_{Weight}), SVTRv2 (37.31% of S_{Mean}, 69.40% of S_{Weight}), and SMTR (35.85% of S_{Mean}, 69.82% of S_{Weight}). This superiority also holds consistently across all 12 specific challenges, where EduOCR achieves the highest PNAcc score in every challenge and the highest 1-NED in 11 out of 12. In particular, it excels in complex challenges such as `Low-quality` (40.16% vs. 33.97%), `Supplement` (25.50% vs. 20.97%), and `Compressed at line end` (15.25% vs. 11.01%). Despite EduOCR's overall excellence, certain complex challenges remain difficult: its performance on `Switching` (PNAcc: 11.31%) and `Compressed at line end` (PNAcc: 15.25%) indicates clear remaining challenges.

C. Test OCR tools/LLMs on BNU-Exam-Benchmark

We evaluate a range of OCR-tools (GOT-OCR2.0 [61], PPOCRv5 [62], TrOCR [63], RapidOCR [64], Tesseract [65], MonkeyOCR [40], dotsOCR [42], DeepSeek-OCR [44], and GLM-OCR [46]) and large language/vision-language models (LLMs/VLMs), including Qwen-VL [66], [67], [68], Qwen3-VL [69], Gemini [70], Kimi [71], [72], Doubao [73], DeepSeek [74], Mistral [75], GPT-4o [76], Claude-4 [77]) on the BNU-Exam-Benchmark. The results detailed in Tables VII and VIII reveal the limitations of current non-specialized approaches. For LLMs/VLMs, the prompt is "*Please output only the text content from the image without any additional descriptions or formatting*" for English, and "请识别图片中的文字内容,不需要任何其他描述或格式" for Chinese.

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

PERFORMANCE COMPARISON ON THE BNU-EXAM-BENCHMARK-ZH. FOR EACH METHOD, THE TOP AND BOTTOM ROWS SHOW PNACC (%) AND 1-NED (%) SCORES, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE. LAST COLUMN SHOWS MODEL PARAMETERS (PARAMS, IN MILLIONS)

Method	comm.	long	ultra	low-qual.	wrong	hard era.	soft era.	suppl.	switch.	repl.	comp.	multi	S_{Mean}	S_{Weight}	Params
CRNN	71.13	47.27	15.29	11.93	16.83	22.18	20.23	1.10	0.00	2.52	0.28	1.79	17.55	48.77	19.73
(TPAMI 2016)	89.79	95.36	92.12	83.33	85.11	87.44	87.42	76.25	76.27	82.46	70.61	71.51	83.14	89.25	
ASTER*	72.98	40.41	14.24	12.75	18.51	23.04	21.32	1.57	0.78	4.53	2.54	2.83	17.96	48.77	20.72
(TPAMI 2019)	86.18	93.98	87.34	78.22	81.07	85.83	86.22	68.27	72.49	77.13	68.93	62.10	79.01	86.12	
NRTR	87.02	49.24	27.05	22.67	30.39	33.90	30.35	5.08	2.10	9.15	8.19	5.67	25.90	60.34	51.32
(ICDAR 2019)	95.27	96.00	93.27	86.16	86.96	90.11	90.60	72.84	77.65	81.99	76.29	65.77	84.42	92.70	
ABINet	84.39	57.27	13.28	24.86	28.21	23.27	27.84	4.62	1.84	13.07	3.67	4.73	23.92	58.86	51.01
(CVPR 2021)	94.80	96.36	91.32	87.59	89.10	88.65	90.19	77.99	77.92	86.28	75.72	73.33	85.77	92.66	
SVTR	87.74	67.35	35.08	28.68	30.49	38.09	40.63	10.99	2.63	18.22	5.36	9.93	31.26	65.82	27.17
(IJCAI 2022)	96.42	98.03	96.58	90.78	91.78	93.28	93.90	86.85	83.32	91.35	81.06	82.95	90.53	95.26	
PARSeq	84.53	29.79	3.84	9.56	18.41	14.19	12.37	2.03	1.05	4.01	3.38	2.08	15.43	50.62	29.26
(ECCV 2022)	94.33	92.53	77.87	74.10	79.77	77.62	81.04	65.02	67.83	71.72	58.81	56.49	74.76	87.79	
LISTER	72.34	34.07	6.88	5.73	11.58	15.05	13.29	0.27	0.00	0.08	0.00	0.47	13.31	45.07	53.92
(ICCV 2023)	90.23	94.27	87.91	79.18	83.18	84.83	85.04	71.64	72.04	77.91	66.96	64.82	79.84	88.17	
SMTR	90.80	70.47	40.91	33.78	40.59	44.43	46.07	13.21	1.84	26.24	7.62	14.09	35.85	69.82	21.11
(AAAI 2025)	96.71	97.54	93.18	90.75	91.99	92.53	93.02	83.95	78.36	90.52	79.57	80.20	89.03	95.01	
IGTR	86.31	62.53	22.08	25.22	28.81	32.50	35.28	9.14	4.21	15.78	3.10	7.85	27.73	62.66	29.68
(TPAMI 2025)	95.45	97.54	93.04	88.98	89.37	91.02	91.63	82.15	78.56	88.08	79.03	77.64	87.71	93.86	
SVTRv2	88.98	71.27	47.51	33.78	37.62	45.15	47.99	17.56	6.57	26.93	7.34	17.12	37.31	69.40	28.98
(ICCV 2025)	96.98	98.30	97.35	92.61	92.79	94.89	95.66	89.94	84.16	93.23	85.13	87.26	92.36	96.18	
SVTRv2-AR	91.09	71.45	44.16	33.97	40.39	46.00	46.73	20.97	6.57	26.76	11.01	18.25	38.11	70.62	28.98
(ICCV 2025)	97.24	96.94	92.76	90.66	91.99	92.72	93.14	87.86	79.90	90.32	79.60	82.06	89.60	95.25	
EduOCR-CTC	89.09	70.74	45.79	33.33	35.64	45.61	45.40	18.29	9.47	25.28	5.64	15.98	36.69	69.23	33.54
	97.19	98.19	97.48	92.41	92.87	94.65	95.13	89.77	85.23	93.35	84.00	86.60	92.24	96.19	
EduOCR	92.80	74.57	53.25	40.16	47.02	52.44	51.00	25.50	11.31	32.08	15.25	22.80	43.18	73.99	33.54
	98.06	98.44	98.11	93.97	94.37	95.68	95.23	91.54	86.09	94.16	87.11	88.29	93.42	97.02	

TABLE VII

PERFORMANCE COMPARISON AGAINST OCR TOOLS AND LARGE/VISION LANGUAGE MODELS (LLMS/VLMS) ON THE BNU-EXAM-BENCHMARK-EN. FOR EACH METHOD, THE TOP AND BOTTOM ROWS SHOW PNACC (%) AND 1-NED (%) SCORES, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Class	Method	comm.	long	ultra	low-qual.	unsp.	hard era.	soft era.	suppl.	switch.	repl.	comp.	multi	S_{Mean}	S_{Weight}
LLM/VLM	Qwen-VL-ocr	82.73	79.93	62.72	53.63	37.09	46.00	40.53	36.15	3.74	43.24	43.69	18.62	45.67	62.54
		95.69	99.38	99.12	95.16	95.04	85.09	77.34	88.48	60.74	94.34	95.03	87.73	89.43	93.38
	Qwen3-VL-plus	82.55	80.25	62.72	53.75	39.68	29.56	17.18	25.59	0.39	30.19	37.24	11.86	39.25	58.15
		95.04	99.36	99.13	94.56	95.38	76.82	68.23	84.69	58.72	87.90	94.60	83.35	86.48	91.17
	Kimi-latest	66.80	46.73	17.75	33.67	25.47	10.83	2.69	11.15	1.38	5.40	24.63	2.31	20.73	32.08
		91.87	98.03	97.18	92.04	93.67	74.48	67.88	82.60	57.96	82.48	92.69	77.95	84.06	88.61
	Doubao-1.5-Vision-Pro	76.32	81.05	63.01	52.61	39.12	42.37	35.86	31.87	8.08	39.04	40.51	15.75	43.79	60.02
		94.05	99.40	99.13	94.50	95.23	83.07	74.83	86.66	62.63	91.82	94.37	85.34	88.41	92.29
	DeepSeek-VL2-base	46.52	41.71	21.35	25.25	17.24	10.90	8.86	9.56	0.98	9.41	20.50	2.50	17.90	27.17
		70.56	90.62	86.24	82.91	85.62	63.03	61.14	76.38	57.98	72.99	88.43	67.66	75.30	76.29
	Gemini-3.1-Pro	77.43	77.86	60.82	51.23	39.12	26.71	10.68	21.71	2.16	16.30	30.79	9.91	35.39	54.41
		92.68	99.27	99.02	94.20	95.72	75.56	67.06	79.80	59.11	81.58	92.80	78.96	84.65	89.82
	Mistral	10.50	42.11	29.20	13.95	8.00	3.55	1.66	1.89	0.00	2.32	14.43	0.37	10.66	17.37
		32.49	90.79	91.72	65.28	64.39	41.74	49.51	62.39	52.50	59.27	76.22	50.86	61.43	61.86
	GPT-4o	71.90	66.40	44.83	40.40	29.53	23.95	19.63	25.99	7.10	24.04	36.66	9.08	33.29	47.26
		91.20	98.67	98.02	91.82	93.00	75.13	70.21	87.11	61.21	88.98	94.54	82.61	86.04	89.46
	Claude-4-sonnet	44.37	49.20	38.47	23.39	14.99	10.19	5.14	10.25	0.39	9.78	20.78	2.68	19.13	31.22
		77.78	97.62	97.34	86.27	85.23	64.52	62.68	80.29	57.46	80.39	89.80	72.15	79.29	82.60
OCR-tools	GOT-OCR2.0	35.96	7.08	1.62	4.38	1.24	7.74	3.64	0.69	0.00	1.86	3.75	0.09	5.67	12.14
		70.80	56.66	52.01	39.75	42.26	38.81	37.12	43.87	42.07	33.48	62.22	27.03	45.50	52.38
	MonkeyOCR-Pro3B	76.34	51.11	22.27	36.98	23.78	14.70	4.43	13.14	0.19	9.51	24.83	2.87	23.35	37.64
		91.76	98.24	97.45	91.99	93.44	72.98	68.03	82.02	58.88	84.55	92.42	77.44	84.10	88.76
	dotsOCR	56.25	69.82	53.88	39.92	28.18	15.25	10.92	14.74	2.36	14.63	26.66	3.79	28.03	43.61
		71.26	97.39	98.06	86.32	89.26	60.45	63.87	76.89	56.78	77.79	88.04	72.70	78.23	80.70
	DeepSeek-OCR	18.08	54.93	34.58	20.20	14.54	7.11	5.77	3.28	0.00	6.33	21.36	0.92	15.59	24.02
		47.82	95.70	96.35	78.17	82.58	57.34	60.92	74.76	56.43	75.99	87.75	68.03	73.49	72.69
	GLM-OCR	73.38	77.07	55.79	46.18	33.48	29.17	12.74	9.86	0.00	15.00	20.40	6.02	31.59	51.89
		88.16	97.16	95.96	89.93	92.72	74.67	65.55	76.97	53.38	80.77	88.26	75.87	81.62	86.98
	PPOCRv5	67.32	10.35	0.14	6.43	0.45	9.72	1.42	0.00	0.00	1.20	0.00	0.00	8.08	20.28
		92.11	77.19	31.27	64.36	48.35	59.79	48.70	29.82	45.42	30.06	26.25	19.32	47.72	60.10
	TrOCR-base-handwritten	21.93	6.60	2.33	5.83	2.48	1.10	0.23	0.29	0.00	0.65	0.96	0.00	3.53	7.52
		75.26	94.50	92.22	82.80	80.51	57.17	57.95	67.89	54.63	69.18	70.91	57.95	71.74	77.56
	Tesseract	1.85	0.07	0.00	0.00	0.00	7.00	10.29	0.00	0.00	0.09	0.00	0.18	1.62	2.29
		15.23	37.92	34.72	33.63	43.28	27.97	35.38	26.36	27.68	16.96	42.59	17.80	29.96	28.39
Our	EduOCR	96.86	90.60	74.25	74.74	56.48	64.42	70.38	39.14	32.34	56.75	18.57	27.06	58.47	77.64
		99.50	99.73	99.46	98.53	97.90	95.58	94.92	92.58	87.01	97.14	88.55	91.76	95.22	98.03

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
PERFORMANCE COMPARISON AGAINST OCR TOOLS AND LARGE/VISION LANGUAGE MODELS (LLMS/VLMS) ON THE BNU-EXAM-BENCHMARK-ZH. FOR EACH METHOD, THE TOP AND BOTTOM ROWS SHOW PNACC (%) AND 1-NED (%) SCORES, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Class	Method	comm.	long	ultra	low-qual.	wrong	hard era.	soft era.	suppl.	switch.	repl.	comp.	multi	S_{Mean}	S_{Weight}
LLM/VLM	Qwen-VL-ocr	86.10	65.56	35.56	28.14	34.65	34.67	34.36	13.86	8.68	21.27	14.68	9.74	32.27	64.14
		96.28	98.00	95.71	89.72	90.38	90.81	89.89	86.56	81.09	91.33	86.58	81.57	89.83	94.59
	Qwen3-VL-plus	81.11	50.04	17.30	19.49	25.34	15.05	14.54	2.21	6.57	7.75	8.47	3.02	20.91	53.85
		93.72	96.33	91.68	83.90	85.93	83.04	83.59	76.56	77.06	83.69	77.51	68.91	83.49	90.69
	Kimi-latest	82.82	60.21	25.71	25.68	30.09	14.89	17.14	2.49	3.94	8.89	3.95	3.40	23.27	57.23
		94.18	97.30	92.71	86.27	86.88	83.51	84.78	80.56	77.30	84.28	75.50	72.37	84.64	91.49
	Doubao-1.5-Vision-Pro	78.97	58.43	28.96	23.04	29.80	24.74	23.49	3.78	6.89	11.85	9.03	5.29	25.36	56.61
		91.71	97.37	94.43	84.94	86.95	87.12	86.29	80.25	76.41	86.40	80.65	73.40	85.49	90.85
	DeepSeek-VL2-base	33.57	7.31	0.86	2.00	4.75	2.40	1.33	0.18	2.10	0.43	0.28	0.09	4.61	18.52
		52.78	54.59	36.89	27.98	37.24	40.61	39.13	27.46	31.63	25.69	25.48	18.43	34.83	47.18
	Gemini-3.1-Pro	67.78	37.55	10.70	12.20	19.80	11.40	12.29	2.31	3.68	6.27	7.06	2.27	16.11	43.73
		87.47	94.90	89.55	79.27	82.23	80.71	82.57	76.55	75.31	82.13	76.67	68.88	81.36	86.55
	Mistral	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.00	0.01	0.07
		9.12	16.82	13.31	4.83	7.97	10.15	10.33	4.85	8.44	5.63	3.82	2.75	8.17	10.21
	GPT-4o	38.06	3.12	0.00	0.72	2.67	1.00	1.08	0.00	0.78	0.26	0.00	0.18	3.99	19.68
		63.73	66.90	45.91	32.45	44.19	42.37	48.18	40.83	42.27	41.13	34.24	34.19	44.70	56.75
	Claude-4-sonnet	17.74	0.80	0.00	0.27	0.69	0.15	0.16	0.00	0.00	0.08	0.00	0.00	1.66	8.99
		44.82	43.45	27.11	17.16	27.42	25.13	30.39	25.41	27.52	27.73	19.37	20.40	27.99	37.85
OCR-tools	GOT-OCR2.0	24.30	1.96	0.00	0.00	1.18	0.54	0.25	0.00	0.00	0.00	0.00	0.00	2.35	12.49
		61.45	53.02	45.08	13.59	33.21	35.65	30.05	30.25	32.06	26.01	28.94	24.46	34.48	49.46
	MonkeyOCR-Pro3B	72.13	44.06	12.81	10.20	17.32	12.57	10.03	2.58	2.63	6.36	6.21	1.70	16.55	46.97
		89.64	95.09	89.00	77.89	81.06	81.19	81.83	77.48	72.73	81.55	74.80	69.25	80.96	87.57
	dotsOCR	51.96	37.10	10.70	5.46	9.70	8.37	5.76	1.20	3.42	3.05	3.10	0.56	11.70	34.45
		70.79	92.94	86.85	60.32	69.14	74.36	74.82	71.15	67.58	73.33	66.68	60.21	72.35	74.85
	DeepSeek-OCR	21.52	14.18	0.86	1.00	3.36	1.70	1.50	0.18	0.79	0.87	0.28	0.19	3.87	13.61
		48.66	79.43	56.21	37.24	50.98	48.40	56.82	54.71	51.61	54.59	42.98	43.64	52.10	53.93
	GLM-OCR	72.55	47.63	13.86	8.19	13.46	12.72	12.45	0.92	1.84	3.05	1.41	0.94	15.75	47.65
		88.40	94.04	83.99	74.88	78.12	78.18	79.90	71.48	70.03	75.77	65.90	66.53	77.27	85.65
	PPOCRv5	77.76	30.24	3.91	3.64	7.92	5.27	3.34	0.18	0.00	0.26	0.00	0.28	11.06	45.14
		93.81	93.36	80.06	64.50	75.93	72.43	76.52	60.15	70.08	65.67	50.09	50.45	71.09	85.85
	RapidOCR	60.29	17.75	2.58	1.36	3.86	1.31	1.92	0.09	0.00	0.00	0.00	0.09	7.74	33.46
		87.38	90.02	80.93	64.80	70.78	71.22	75.28	57.43	65.17	67.10	48.44	48.55	68.93	81.82
	Tesseract	0.42	0.00	0.00	0.00	0.00	2.17	2.67	0.00	0.00	0.00	0.00	0.00	0.43	0.61
		7.02	8.61	6.44	3.40	5.47	10.27	8.64	3.48	5.29	4.13	3.20	3.04	5.75	7.34
Our	EduOCR	92.80	74.57	53.25	40.16	47.02	52.44	51.00	25.50	11.31	32.08	15.25	22.80	43.18	73.99
		98.06	98.44	98.11	93.97	94.37	95.68	95.23	91.54	86.09	94.16	87.11	88.29	93.42	97.02

Among the LLMs/VLMs, some recent models like Qwen-VL-OCR demonstrate foundational capabilities, achieving the highest SWeight PNAcc scores in their class with 62.54% on the English and 64.14% on the Chinese benchmarks. However, a significant performance disparity exists, with many prominent models performing poorly, especially on the more complex Chinese handwriting task. Similarly, most OCR tools, including Tesseract, are not equipped for the benchmark's challenges, yielding near-zero accuracy. While some tools like PPOCRv5 show moderate success on the Chinese subset with an SWeight score of 45.14%, they still struggle with diverse handwriting artifacts and layout irregularities. Large-scale OCR models, such as MonkeyOCR, dotsOCR, DeepSeek-OCR, and GLM-OCR, perform better than traditional OCR tools in several cases, but they remain clearly below EduOCR on both English and Chinese, especially on artifact-heavy categories. This comprehensive evaluation indicates that both LLM-based and OCR tools currently lack the robustness and high fidelity required to reliably process exam HTR. Fig. 10 shows the visualization results. It is worth noting that LLMs/VLMs tend to conduct "semantic correction", automatically rectifying students' writing errors and thereby reducing fidelity to the original writing.

D. Visual Analysis

The category-level results show that `Supplement` and `Switching` are challenging because the physical writing layout may differ from the intended logical reading order. To further examine how EduOCR handles such cases, we visualize the cross-attention maps of the Sequential Symbol Module (SSM) decoder in Fig. 11. For `Supplement`, EduOCR does not simply attend along the main text line. In both English and Chinese examples, the attention first follows the main-line text, then shifts to the inserted region, and finally returns to the main line. This indicates that the model can adjust its decoding path according to the supplement structure. For `Switching`, the attention changes mainly around the correction mark. Instead of strictly following the original physical order, the decoder locally adjusts the attention sequence according to the switching cue. These visualizations show that EduOCR can adapt its attention trajectory to correction structures. The decoder can switch between main text, inserted text, and locally reordered regions, producing a sequence closer to the annotated logical reading order. This observation is consistent with the quantitative improvements of EduOCR on order-disrupting artifacts.

E. Ablation Study

To validate the effectiveness of the proposed modules and analyze model scalability, we conduct a series of ablation studies on the BNU-Exam-Benchmark-EN. Table IX reports module ablations and the direct comparison between PPH and the feature rearrangement module (FRM) of SVTRv2, while Table X reports scalability analysis.

Effect of Symbol Awareness Block (SAB): SAB is designed to handle artifact symbols with extreme aspect ratios by expanding the horizontal receptive field via depth-wise separable convolutions. Its contribution is assessed by comparing EduOCR with

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.



Fig. 10. Qualitative comparison with representative OCR tools and LLMs/VLMs on BNU-Exam-Benchmark. Total line correct predictions are in green; errors are in red.

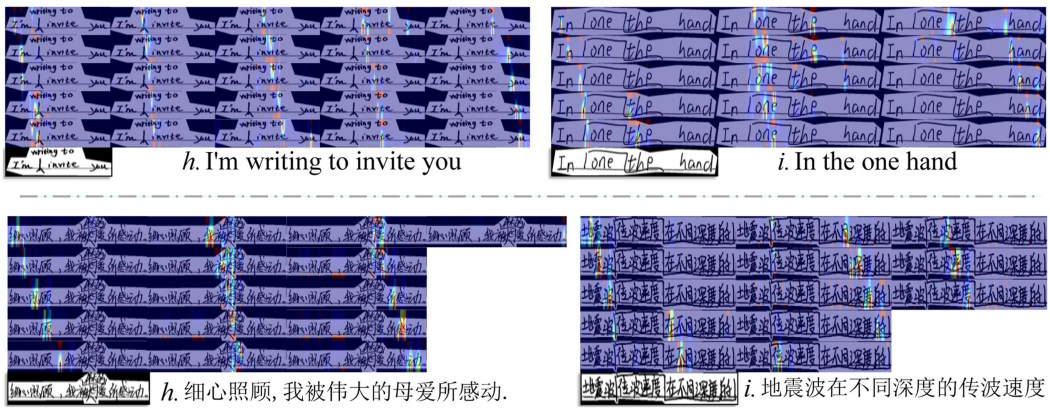


Fig. 11. Visual analysis of EduOCR attention on the BNU-Exam-Benchmark. In `Supplement`, the decoder shifts between the main text line and the inserted region. In `Switching`, it locally adjusts the attention order according to the correction mark. .

a variant without SAB (denoted as "w/o SAB"). As shown in Table IX, removing SAB leads to a slight drop: S_{Weight} decreases from 77.64% to 76.19%, and S_{Mean} drops from 58.47% to 56.70%. The degradation is more pronounced in artifact-related scenarios, such as `Supplement` (39.14% vs. 36.13%) and `Switching` (32.34% vs. 24.45%). This indicates that SAB effectively enhances the encoder's ability to disentangle artifact symbols from image, supporting robust feature extraction in exam scenarios.

Effect of Permutation-Aware Prediction Head (PPH): PPH regularizes the shared encoder through permutation-biased and noise-perturbed queries. As shown in Table IX, removing PPH decreases $S_{\text{Mean}}/S_{\text{Weight}}$ from 58.47%/77.64% to 54.96%/76.10%, with clear drops on `Supplement` (39.14% vs. 33.46%) and `Switching` (32.34% vs. 18.73%). A further variant replaces PPH with SVTRv2's feature rearrangement module (FRM), denoted as +FRM. FRM is designed to canonicalize a geometrically distorted but continuous text trajectory into a CTC-compatible sequence, whereas exam artifacts introduce correction-induced non-monotonicity. Its column-wise rearrangement may compress competing cues into a single token and still follows a monotonic CTC path. In contrast, PPH provides softer perturbation-based supervision, while SSM reconstructs the final non-monotonic order. +FRM obtains 55.16%/75.73%,

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
ABLATION COMPARISON ON THE BNU-EXAM-BENCHMARK-EN. WE EVALUATE VARIANTS OF OUR MODEL WITHOUT THE SYMBOL AWARENESS BLOCK (W/O SAB), THE PERMUTATION-AWARE PREDICTION HEAD (W/O PPH), THE SEQUENTIAL SYMBOL MODULE (W/O SSM), AND A VARIANT THAT REPLACES PPH WITH SVTRV2'S FEATURE REARRANGEMENT MODULE (+FRM)

Method	comm.	long	ultra	low-qual.	unsp.	hard era.	soft era.	suppl.	switch.	repl.	comp.	multi	S_{Mean}	S_{Weight}	Params
w/o SAB	96.02	90.44	72.41	74.68	54.34	63.79	70.30	36.13	24.45	56.01	16.61	25.30	56.70	76.19	25.15
	99.53	99.68	99.12	98.47	97.73	95.33	94.42	92.10	84.32	97.16	89.25	91.36	94.87	97.75	
w/o PPH	96.75	90.04	72.20	73.30	51.18	63.63	66.27	33.46	18.73	53.12	17.90	22.89	54.96	76.10	25.25
	99.48	99.71	99.39	98.32	97.34	95.42	91.76	90.83	81.10	96.35	87.68	90.65	94.00	97.61	
+FRM	96.50	89.49	72.63	73.60	52.31	60.07	66.74	35.35	18.54	52.93	19.44	24.37	55.16	75.73	27.35
	99.42	99.69	99.41	98.33	97.48	93.46	92.32	91.86	83.07	96.55	88.99	91.08	94.31	97.43	
w/o SSM	95.41	90.12	68.38	67.46	41.26	52.64	59.69	17.72	19.92	43.24	5.38	5.38	48.06	71.61	18.41
	99.25	99.74	99.28	98.14	96.78	92.48	90.93	89.99	86.00	96.02	85.89	89.20	93.65	96.97	
EduOCR	96.86	90.60	74.25	74.74	56.48	64.42	70.38	39.14	32.34	56.75	18.57	27.06	58.47	77.64	25.58
	99.50	99.73	99.46	98.53	97.90	95.58	94.92	92.58	87.01	97.14	88.55	91.76	95.22	98.03	

TABLE X
MODEL SCALABILITY COMPARISON OF EDUOCR VARIANTS ON THE BNU-EXAM-BENCHMARK-EN

Method	dims	depths	mixers	comm.	long	ultra	low-qual.	unsp.	hard era.	soft era.	suppl.	switch.	repl.	comp.	multi	S_{Mean}	S_{Weight}	Params
EduOCR-T	[64,128,256]	[3,6,3]	$[S]_6[G]_6$	96.86	88.93	70.86	73.54	49.37	60.47	65.24	30.47	23.07	49.48	16.07	22.79	53.93	74.93	7.70
				99.51	99.66	99.37	98.29	97.19	93.95	93.39	90.54	80.59	95.65	87.42	90.51	93.84	97.51	
EduOCR-S	[96,192,256]	[3,6,6]	$[S]_8[G]_8$	96.86	89.80	74.32	74.02	52.87	60.86	69.27	35.95	26.62	55.26	17.80	23.91	56.47	76.66	12.14
				99.48	99.72	99.44	98.57	97.65	94.17	94.47	92.05	84.78	96.69	88.45	91.34	94.73	97.76	
EduOCR-B	[128,256,384]	[6,6,6]	$[S]_8[G]_{10}$	96.77	91.16	73.90	75.52	54.11	63.08	70.22	37.94	30.57	55.63	17.90	25.85	57.72	77.31	25.71
				99.83	99.90	99.72	98.74	98.70	96.56	95.20	93.63	87.89	97.48	89.77	93.09	95.88	98.46	

lower than EduOCR's 58.47%/77.64%, confirming the benefit of PPH for correction-induced irregularity.

Effect of Sequential Symbol Module (SSM): SSM is the primary autoregressive decoder responsible for precise sequence generation. Its necessity is evaluated using a variant without SSM ("w/o SSM"). As expected, removing SSM causes a clear performance decline: S_{Weight} decreases from 77.64% to 71.61%, and S_{Mean} drops from 58.47% to 48.06%. The degradation is concentrated on complex challenges such as `Supplement` (39.14% vs. 17.72%), `Replacement` (56.75% vs. 43.24%), `Compressed at line end` (18.57% vs. 5.38%), and `Multi-Type` (27.06% vs. 5.38%). This highlights SSM's core role in modeling long-range dependencies and reconstructing the intended reading order, which is indispensable for handling nonlinear layouts in exams.

Model Scalability: Model scalability is analyzed using three EduOCR variants with increasing capacity: EduOCR-T (7.70 M), EduOCR-S (12.14 M), and EduOCR-B (25.71 M). As shown in Table X, overall performance improves with model size: S_{Mean}/S_{Weight} increases from 53.93%/74.93% (T) to 56.47%/76.66% (S) and 57.72%/77.31% (B). The gain is clear on challenging categories such as `Supplement` (30.47% to 37.94%) and `Switching` (23.07% to 30.57%). Even the tiny variant already outperforms CRNN on S_{Weight} (74.93% vs. 52.30%), showing that the proposed architecture is effective even under a smaller model scale. The remaining gap between EduOCR-S and EduOCR-B on the hardest subsets also suggests that exam HTR is still capacity-demanding and can benefit from further scaling.

F. Discussion

The results of this study highlight the distinctive nature and broader significance of handwritten text recognition (HTR) in examination scenarios. Unlike conventional HTR or scene text recognition (STR), which emphasize semantic accuracy or readability, exam HTR demands verbatim transcription fidelity, since any alteration of characters, such as omission, rewriting, or semantic correction, can directly influence grading outcomes. This requirement shifts the objective of recognition from semantic interpretation to the faithful reproduction of student handwriting, establishing exam HTR as a distinct and specialized subdomain.

1) Insights From Dataset and Benchmark: The proposed BNU-Exam-HTR dataset and BNU-Exam-Benchmark highlight how the complexity of real exam handwriting extends far beyond conventional datasets. The inclusion of 12 challenge types, covering artifacts such as erasure, supplement, switching, and replacement, exposes the insufficiency of current methods to handle irregular visual and sequential disruptions. Empirical evidence from Section V shows that existing models maintain high 1-NED but fail drastically in PNAcc, particularly under artifact-heavy categories. This indicates that current architectures, though strong in average accuracy, lack robustness in educationally critical cases. The benchmark thus provides not only a rigorous testing bed but also a conceptual framework for phenomenon-aware evaluation, enabling future research to pinpoint model weaknesses in a structured manner.

2) Architectural Implications: The proposed EduOCR model demonstrates that improving recognition in exam settings requires jointly modeling visual and sequential irregularities. Its dual-branch architecture, which includes the Sequential Symbol Module (SSM) and the Permutation-Aware Prediction Head (PPH), illustrates how separating robust feature learning from non-monotonic sequence decoding can effectively mitigate alignment drift and semantic hallucination. The strong performance of EduOCR on both English and Chinese subsets suggests that such collaborative mechanisms may serve as a general design principle for artifact-resilient recognition models. Beyond exam papers, this insight could

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

inform document OCR, handwritten forms, and educational note recognition, where overlapping, correction, and spatial reordering phenomena are also common.

3) Limitations: Despite the comprehensive design, several limitations remain. At the dataset level, BNU-Exam-HTR focuses on English and Chinese text lines and excludes formulas, tables, diagrams, page-level mixed layouts, and entirely illegible regions. Future datasets should therefore extend to page-level exam-document understanding, covering mixed printed-handwritten layouts, formulas, tables, figures, and reading-order relations. At the model level, EduOCR is a line-level recognizer and still lacks explicit inter-line and hierarchical spatial reasoning. Extending the current framework to page-level exam document understanding would require additional modules for layout analysis and region routing, as well as specialized decoders for formulas, tables, chemical structures, and figures. EduOCR also remains weak on Chinese `Switching` (11.31%) and `Compressed at line end` (15.25%), suggesting future work on relation-aware correction modeling and local high-resolution refinement.

VI. Conclusion

This paper investigates the challenges of handwritten text recognition in examination scenarios. We present BNU-Exam-HTR, the largest dataset of its kind, and introduce BNU-Exam-Benchmark for fine-grained, challenge-specific evaluation. Building upon this benchmark, we develop EduOCR, a dual-branch recognition model that demonstrates strong robustness and adaptability to diverse handwriting artifacts. Experimental results confirm its state-of-the-art performance, surpassing existing HTR models. The proposed dataset, benchmark, and model together provide a solid foundation for studies on exam-oriented HTR and intelligent grading systems.

In our future work, we plan to enhance spatial understanding, multimodal integration, and educational applicability in exam HTR. We aim to develop models that better capture the spatial layout of handwritten answers and integrate visual, textual, and contextual cues for more coherent reasoning. Furthermore, we will focus on improving robustness and fairness across diverse handwriting styles and examination contexts, moving toward a more human-centered and artifact-aware handwriting recognition framework that bridges visual perception and educational cognition.

References

[1] W. Yang, H. Zhang, and J. Lin, "Simple applications of BERT for ad hoc document retrieval," Mar. 2019, *arXiv:1903.10972*. .

[2] K. Maekawa et al., "Improving OCR for historical documents by modeling image distortion," in *Proc. Int. Conf. Asian Digit. Libraries*, Jan. 2019, pp. 312–316.

[3] H. Ren et al., "ST-SiameseNet: Spatio-temporal siamese networks for human mobility signature identification," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discov.; Data Mining*, Aug. 2020, pp. 1306–1315.

[4] J. Poddar, V. Parikh, and S. K. Bharti, "Offline signature recognition and forgery detection using deep learning," *Procedia Comput. Sci.*, vol. 170, pp. 610–617, Jan. 2020, doi: 10.1016/j.procs.2020.03.133.

[5] A. Sharma and D. B. Jayagopi, "Automated grading of handwritten essays," in *Proc. 16th Int. Conf. Front. Handwriting Recognit.*, 2018, pp. 279–284.

[6] E. Shaikh et al., "Automated grading for handwritten answer sheets using convolutional neural networks," in *Proc. 2nd Int. Conf. New Trends Comput. Sci.*, 2019, pp. 1–6.

[7] M. N. I. Susanti, A. Ramadhan, and H. L. H. S. Warnars, "Automatic essay exam scoring system: A systematic literature review," *Procedia Comput. Sci.*, vol. 216, pp. 531–538, 2023.

[8] H. T. Nguyen et al., "Handwriting recognition and automatic scoring for descriptive answers in Japanese language tests," in *Proc. Int. Conf. Front. Handwriting Recognit.*, Jan. 2022, pp. 274–284.

[9] Y. Zhu, Z. Xie, L. Jin, X. Chen, Y. Huang, and M. Zhang, "SCUT-EPT: A new dataset and benchmark for offline Chinese text recognition in examination paper," *IEEE Access*, vol. 7, pp. 370–382, 2019.

[10] T. Team, 2020. [Online]. Available: https://ai.100tal.com/openData/ocr

[11] A. Graves et al., "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *Proc. 23rd Int. Conf. Mach. Learn.*, 2006, pp. 369–376.

[12] B. Shi, X. Bai, and C. Yao, "An end-to-end trainable neural network for image-based sequence recognition and its application to scene text recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 11, pp. 2298–2304, Nov. 2017, doi: 10.1109/TPAMI.2016.2646371.

[13] Y. Du et al., "SVTR: Scene text recognition with a single visual model," in *Proc. 31st Int. Joint Conf. Artif. Intell.*, 2022, pp. 884–890.

[14] B. Shi, M. Yang, X. Wang, P. Lyu, C. Yao, and X. Bai, "ASTER: An attentional scene text recognizer with flexible rectification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 9, pp. 2035–2048, Sep. 2019, doi: 10.1109/TPAMI.2018.2848939.

[15] F. Sheng, Z. Chen, and B. Xu, "NRTR: A no-recurrence sequence-to-sequence model for scene text recognition," in *Proc. Int. Conf. Document Anal. Recognit.*, Sep. 2019, pp. 781–786, doi: 10.1109/icdar.2019.00130.

[16] H. Li et al., "Show, attend and read: A simple and strong baseline for irregular text recognition," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 8610–8617.

[17] D. Bautista and R. Atienza, "Scene text recognition with permuted autoregressive sequence models," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 178–196.

[18] A. Hannun, "Sequence modeling with CTC," *Distill*, vol. 2, no. 11, 2017, Art. no. e8.

[19] U.-V. Marti and H. Bunke, "The IAM-database: An english sentence database for offline handwriting recognition," *Int. J. Document Anal. Recognit.*, vol. 5, pp. 39–46, Nov. 2002, doi: 10.1007/s100320200071.

[20] P. Grother and K. Hanaoka, "NIST Special Database 19: Handprinted Forms and Characters Database," National Institute of Standards and Technology, 2016.

[21] P. Krishnan, R. Kovvuri, G. Pang, B. Vassilev, and T. Hassner, "TextStyleBrush: Transfer of text aesthetics from a single example," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 9122–9134, Jul. 2023.

[22] A. W. C. Lee, J. Chung, and M. Lee, "GNHK: A dataset for English handwriting in the wild," in *Proc. Int. Conf. Document Anal. Recognit.*, 2021, pp. 399–412.

[23] P. Hu et al., "Handwritten style recognition for chinese characters on HCL2020 dataset," in *Proc. 3rd Chin. Conf. Pattern Recognit. Comput. Vis.*, Nanjing, China, 2020, pp. 138–150.

[24] C.-L. Liu, F. Yin, D. -H. Wang, and Q. -F. Wang, "CASIA online and offline Chinese handwriting databases," in *Proc. Int. Conf. Document Anal. Recognit.*, 2011, pp. 37–41.

[25] H. Zhang, L. Liang, and L. Jin, "SCUT-HCCDoc: A new benchmark dataset of handwritten Chinese text in unconstrained camera-captured documents," *Pattern Recognit.*, vol. 108, 2020, Art. no. 107559.

[26] R. Plamondon and S. N. Srihari, "Online and off-line handwriting recognition: A comprehensive survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 1, pp. 63–84, Jan. 2000.

[27] D. Zhong et al., "Struck-out handwritten word detection and restoration for automatic descriptive answer evaluation," *Signal Process.: Image Commun.*, vol. 130, 2025, Art. no. 117214.

[28] C. Neudeckeret al., "A survey of OCR evaluation tools and metrics," in *Proc. 6th Int. Workshop Historical Document Imag. Process.*, New York, NY, USA, Association for Computing Machinery, 2021, pp. 13–18, doi: 10.1145/3476887.3476888.

[29] N. Ezaki et al., "Off-line recognition of realistic Chinese handwriting using segmentation-free strategy," *Pattern Recognit.* vol. 42, no. 1, pp. 167–182, 2009.

[30] F. Yin, Q.-F. Wang, X.-Y. Zhang, and C.-L. Liu, "ICDAR 2013 Chinese handwriting recognition competition," in *Proc. 12th Int. Conf. Document Anal. Recognit.*, 2013, pp. 1464–1470.

[31] Q. Jiang, J. Wang, D. Peng, C. Liu, and L. Jin, "Revisiting scene text recognition: A data perspective," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 20486–20497.

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.

[32] Y. Liuet al., "OCRBench: On the hidden mystery of OCR in large multimodal models," *Sci. China Inf. Sci.*, vol. 67, no. 12, Dec. 2024, Art. no. 220102, doi: 10.1007/s11432-024-4235-6.
[33] Y. Du et al., "Context perception parallel decoder for scene text recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 6, pp. 4668–4683, Jun. 2025.
[34] T. Guan, W. Shen, X. Yang, Q. Feng, Z. Jiang, and X. Yang, "Self-supervised character-to-character distillation for text recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 19416–19427.
[35] T. Guan et al., "Self-supervised implicit glyph attention for text recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 15285–15294.
[36] B. Na, Y. Kim, and S. Park, "Multi-modal text recognition networks: Interactive enhancements between visual and semantic features," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 446–463.
[37] D. Coquenet, C. Chatelain, and T. Paquet, "DAN: A segmentation-free document attention network for handwritten document recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8227–8243, Jul. 2023.
[38] D. Peng et al., "Recognition of handwritten Chinese text by segmentation: A segment-annotation-free approach," *IEEE Trans. Multimedia*, vol. 25, pp. 2368–2381, 2023.
[39] M.-M. Yu et al., "An approach for handwritten chinese text recognition unifying character segmentation and recognition," *Pattern Recognit.*, vol. 151, 2024, Art. no. 110373.
[40] Z. Li et al., "MonkeyOCR: Document parsing with a structure-recognition-relation triplet paradigm," 2025, *arXiv:2506.05218*.
[41] J. Zhang et al., "MonkeyOCR v1. 5 technical report: Unlocking robust document parsing for complex patterns," 2025, *arXiv:2511.10390*.
[42] Y. Li et al., "dots.ocr: Multilingual document layout parsing in a single vision-language model," 2025, *arXiv:2512.02498*.
[43] H. Zheng et al., "Multimodal OCR: Parse anything from documents," 2026, *arXiv:2603.13032*.
[44] H. Wei, Y. Sun, and Y. Li, "DeepSeek-OCR: Contexts optical compression," 2025, *arXiv:2510.18234*.
[45] H. Wei, Y. Sun, and Y. Li, "DeepSeek-OCR 2: Visual causal flow," 2026, *arXiv:2601.20552*.
[46] S. Duan et al., "GLM-OCR technical report," 2026, *arXiv:2603.10910*.
[47] H. V. Team et al., "Hunyuanocr technical report," 2025, *arXiv:2511.19575*.
[48] C. Cheng, P. Wang, C. Da, Q. Zheng, and C. Yao, "LISTER: Neighbor decoding for length-insensitive scene text recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 19484–19494.
[49] Y. Du et al., "Out of length text recognition with sub-string matching," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 3, Apr. 2025, pp. 2798–2806. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/32285
[50] J. Chen et al., "Benchmarking Chinese text recognition: Datasets, baselines, and an empirical study," 2021, *arXiv:2112.15093*.
[51] A. Marzal and E. Vidal, "Computation of normalized edit distance and applications," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 15, no. 9, pp. 926–932, Sep. 1993.
[52] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, pp. 5998–6008, 2017.
[53] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.
[54] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. 14th Int. Conf. Artif. Intell. Statist. JMLR Workshop Conf. Proc.*, 2011, pp. 315–323.
[55] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *Proc. Int. Conf. Learn. Representations*, 2019.
[56] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. Int. Conf. Learn. Representations*, Aug. 2016. pp. 1–16.
[57] Y. Du, Z. Chen, H. Xie, C. Jia, and Y. -G. Jiang, "SVTRv2: CTC beats encoder-decoder models in scene text recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2025, pp. 20147–20156.
[58] S. Fang, H. Xie, Y. Wang, Z. Mao, and Y. Zhang, "Read like humans: Autonomous, bidirectional and iterative language modeling for scene text recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 7094–7103, doi: 10.1109/cvpr46437.2021.00702.
[59] Y. Du, Z. Chen, Y. Su, C. Jia, and Y.-G. Jiang, "Instruction-guided scene text recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2723–2738, Apr. 2025.
[60] F. L. Bookstein, "Principal warps: Thin-plate splines and the decomposition of deformations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no. 6, pp. 567–585, May 1989, doi: 10.1109/34.24792.
[61] H. Wei et al., "General OCR theory: Towards OCR-2.0 via a unified end-to-end model," 2024, *arXiv:2409.01704*.
[62] P. Authors, "PaddleOCR, awesome multilingual OCR toolkits based on paddlepaddle," 2020. [Online]. Available: https://github.com/PaddlePaddle/PaddleOCR
[63] M. Li et al., "TrOCR: Transformer-based optical character recognition with pre-trained models," in *Proc. AAAI Conf. Artif. Intell.*, 2021. pp. 13094–13102.
[64] R. Team, "Rapid OCR: OCR toolbox," 2021. [Online]. Available: https://github.com/RapidAI/RapidOCR
[65] R. Smith, "An Overview of the Tesseract OCR engine," in *Proc. Int. Conf. Document Anal. Recognit.*, vol. 2, 2007, pp. 629–633.
[66] J. Bai et al., "Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond," 2023, *arXiv:2308.12966*.
[67] P. Wang et al., "Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution," 2024, *arXiv:2409.12191*.
[68] S. Bai et al., "Qwen2.5-VL technical report," 2025, *arXiv:2502.13923*.
[69] S. Bai et al., "Qwen3-VL technical report," 2025, *arXiv:2511.21631*.
[70] G. Team et al., "Gemini: A family of highly capable multimodal models," 2023, *arXiv:2312.11805*.
[71] G. Kim et al., "OCR-free document understanding transformer," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 498–517.
[72] K. Team et al., "Kimi-VL technical report," 2025, *arXiv:2504.07491*.
[73] ByteDance, "Doubao large model family," Technology Launch, Volcano Engine Blog, 2024. [Online]. Available: https://seed.bytedance.com/en/research
[74] Z. Wu et al., "DeepSeek-VL2: Mixture-of-experts vision-language models for advanced multimodal understanding," 2024, *arXiv:2412.10302*.
[75] Mistral, "Mistral OCR," Technology Launch, Volcano Engine Blog, 2025. [Online]. Available: https://mistral.ai/news/mistral-ocr
[76] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.
[77] Anthropic, "Claude sonnet 4: A hybrid-reasoning, mid-size model optimized for coding and high-throughput tasks," Anthropic News, May 2025. Accessed: Aug. 11, 2025. [Online]. Available: https://www.anthropic.com/news/claude-4



Runrui Li received the BS and MS degrees from the Taiyuan University of Technology, Taiyuan, China, in 2020 and 2023, respectively. He is currently working toward the PhD degree with the School of Artificial Intelligence, Beijing Normal University, Beijing, China. His research interests include computer vision and handwritten text recognition.



Lin Zhu (Member, IEEE) received the BS degree from Northwestern Polytechnical University, China, in 2014, the MS degree from North Automatic Control Technology Institute, China, in 2018, and the PhD degree from the School of Computer Science, Peking University, China, in 2022. He is currently an associate professor with the School of Artificial Intelligence, Beijing Normal University. His research interests include AI4Education, video processing, and pattern recognition.



Hua Huang (Senior Member, IEEE) received the BS, MS, and PhD degrees from Xi'an Jiaotong University, Xian, China, in 1996, 2001, and 2006, respectively. He is currently a professor with the School of Artificial Intelligence, Beijing Normal University. His research interests include image and video processing, and pattern recognition.

Authorized licensed use limited to: Beijing Normal University. Downloaded on November 12,2026 at 06:54:42 UTC from IEEE Xplore. Restrictions apply.